What does the Super Mario 64 courtyard plaque say?

A forward-model reconstruction of the "L is real 2401" texture: font, process and text

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Abstract

The stone plaque on the star statue in the castle courtyard of *Super Mario 64* (1996) has fuelled one of the longest-running legends in gaming: that it reads "L is real 2401". The plaque is a single 32×32 grey texture with two lines of text about three to five pixels tall, which is far below the resolution needed to read letters by eye. Instead of enhancing or interpreting the image, we *simulate* how a 1996 artist would have produced it. We type a candidate text in a real period font, resize it into the texture with a standard filter, and measure how well the result matches the stored pixels after removing the stone background. We do this for millions of candidate texts, fonts and process settings on a GPU, and calibrate the method on synthetic plaques with known text. We find that (i) the text is set in **Palatino Bold Italic**, most likely Adobe's own 1990 PostScript Type 1 version as used with Adobe Type Manager; it beats every other era font family under an identical, exhaustive process search, including the game's own dialog font, which is clearly ruled out; (ii) a *single rigid designer pipeline* reproduces the texture: with Adobe's file 192×160 artwork, 26 px type, bilinear resize, light blur and gamma 1.4; the Apple and URW Palatinos independently converge on 160×128, 23 px and gamma 1.8. A second pass against the extracted plain stone, an exhaustive test of every 3–4-letter string and 470 era fonts (including everything an SGI workstation could install) changes nothing; working like the designer (hand kerning, a 1 % narrower 190×160 artwork) brings the model to within about one grey level of the plaque, and the few pixels left look hand-retouched; (iii) line 1 reads **"Eternal Star"**: "Star" is the best of 27,122 words as the second word (whether the s is upper or lower case cannot be decided); (iv) line 2 is set in the same process at 17–18 px, centred under line 1, and "hope" is the best of 12,556 middle words. We then fitted line 2's own typesetting on the four letters of "hope" alone. On their own, held-out pixels, "We" becomes the best first word and "so" the best real ending, and the unconstrained dictionary search returns "We hope so" among its top three strings; (v) line 2 is in the *same* font: fitted alone against 3,396 era faces and finally drawn hinted at its true pixel size, as 1996 software drew it, it is again best explained by Palatino Bold Italic, and in the final hand-kerned fit Adobe's file wins on both lines. With Adobe's original Palatino, a blind search over 17.7k words returns exactly "We hope so" as its best string. Our most likely reading is therefore

Eternal Star
We hope so

"L is real 2401", "A Secret star / lie here ~", "My hope is" and other published readings fit the pixels substantially worse under the same model. All code, results and figures are released for independent verification; no Nintendo data is redistributed.

Contents

1 Introduction 3

2 The data 4

3 Method: a forward model of the designer's process 5

4 Identifying the font 6

5 Reconstructing the process 11

6 Line 1: "Eternal Star" 13

7 Line 2: "We hope so" 14

8 Working backwards from "hope" 25

9 Calibration: how much can the pixels tell? 31

10 The artist's font file: Adobe's 1990 PostScript Palatino **33**

11 Inverting the process: can the text be made visible? **36**

12 Second pass: the plain stone, every string and every era font **37**

13 Third pass: thinking like the designer **40**

14 Fourth pass: is line 2 in the same font? And the final fit **45**

15 Comparison with published readings **46**

16 Does the reading make sense in Japanese? **49**

17 Limitations **49**

18 Conclusion **50**

A Reproducing the results **50**



Figure 1: The result in one picture. Top left: the real plaque, a 32×32 texture from the game's ROM. Top right: our best model of how it was made. Bottom: the artwork that model implies, before it was shrunk: "Eternal Star / We hope so" typed in Adobe Palatino Bold Italic (1990) at 26 px and 19 px on one 190×160 canvas, a few letters nudged by hand, then resized to 32×32 with a bilinear filter, light blur and gamma 1.4 onto the stone. Inside the text the model is on average 1.1 grey levels (of 31) from the real plaque (Section 14).

1 Introduction

In the courtyard behind Princess Peach's castle stands a statue topped by a star, with a small engraved plaque at its base (Fig. 2). The plaque is blurry, and for almost three decades players have argued about what it says.

- **"L is real 2401".** This is the classic reading from the late 1990s, taken as a hint that Luigi was hidden in the game [1, 2]. A variant spells the number as "2041" [3].
- **The 1998 Nintendo letter.** A Nintendo of America game-play counsellor's letter from 1998, which surfaced in 2016, called the plaque a developers' joke without saying what it reads [4, 5].
- **"Eternal Star".** This is the reading the Super Mario Wiki considers "more likely", set "in a heavily serifed font" [1].
- **"A Secret star / lie here ~".** Daniel Nodd (No64DD) proposed this in 2015 from photographs of CRT screens, and Nintendo Life covered it in 2019 [6].
- **"My hope is".** A further suggestion for line 2, based on two descenders and an ascender.

All previous attempts share one weakness. They *read* a 32×32 image, whether by eye, by upscaling or by photographing a CRT, and people then disagree about which shapes they see. At this resolution a human (or an upscaler) will find whatever text they are primed to find. This report takes the opposite approach.

1. **Model, don't read.** We write down how such a texture is made: type, resize, blur, gamma, on top of a stone background.
2. **Score hypotheses.** Every candidate (font, process, text) is rendered through that model and compared with the stored pixels.
3. **Calibrate.** Synthetic plaques with known text show what the method can and cannot recover.

We only use the texture as it ships in the game. Leaked source material is not used.

2 The data

The plaque is the texture `castle_grounds_textures.08800`. It is a 32×32 RGBA16 texture inside the MIO0-compressed `outside` texture segment, mapped onto a quad of about 142×105 world units in the courtyard model.

- We extracted it from the US, Japanese, European and Shindou ROMs, using segment offsets from the public decompilation [7].
- It is **byte-identical in all versions** (SHA-1 `88ef618a`…), so the text was never localised or revised.
- The texture is pure grey and uses only 23 distinct 5-bit levels.
  There are two lines of text:
- **line 1:** rows 8–12, columns 2–29;
- **line 2:** mainly rows 17–19, columns 7–23;
- the lower rows are clean stone.

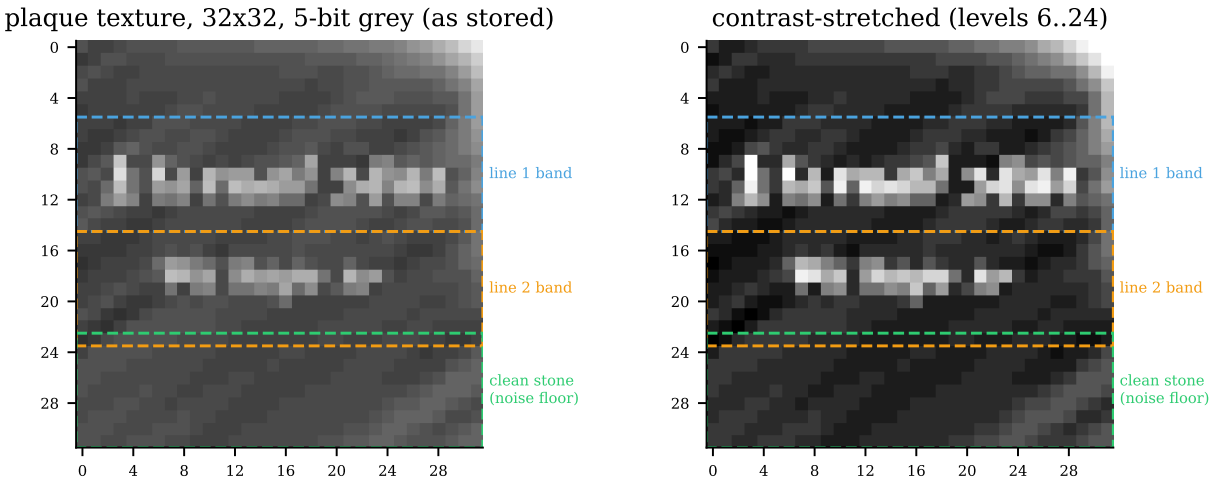


Figure 2: The plaque texture as stored in the ROM (left) and contrast-stretched (right). Dashed boxes mark the two text bands we model and the clean stone rows that define the noise floor.

3 Method: a forward model of the designer's process

Why direct reading fails. At 1.5–2 texture pixels per character, many different strings produce almost the same blur. In early experiments we let every letter move and change freely. Meaningless strings then fitted *as well as or better than* any real sentence, which shows that pixel evidence alone cannot identify the text. Identification only becomes possible once the *process* is constrained by the way the texture was actually made.

Generative model. We model the stored texture t as

$$t \;=\; \text{stone} \;+\; L \cdot \big(\mathcal{R}(\text{text})\big)^{\gamma} \;+\; \text{noise},$$

where $\mathcal{R}$ is the designer's rendering process (Fig. 3):

1. The text is typed *once*, anti-aliased, with font F at an integer pixel size P, with tracking, optional pair kerning, and fractional or whole-pixel advances.
2. It is typed on an artwork canvas of $32k_x \times 32k_y$ pixels.
3. The canvas is resized to 32×32 with a standard filter (nearest, box, bilinear, bicubic, Lanczos), whose support is scaled with the downsampling factor as in Photoshop and PIL.
4. An optional Gaussian blur is applied, then a gamma γ.

The engraving brightness L is solved in closed form.

The stone is a nuisance. Inside each 4-column block of a text band it is modelled as a plane $(a + bx + cy)$ and projected out exactly, so the score is the residual sum of squares (SSE) after the best plane and the best ink level.

Honest noise floor. Applying the identical operator to the nine clean stone rows 23–31 gives SSE = 165.7. This is what pure stone texture with no text costs. No model can be expected to go far below a few times this value, because the stone under the letters is unknown.

Search. All renderings are expressed as matrix products: canvas→texture weight matrices, with every integer position as a slice. This makes exhaustive searches feasible on one consumer GPU (RTX 2060 S): millions of configurations per second for line 1, and a K-best dynamic programme over word sequences for line 2 (Section 7).



Figure 3: The designer pipeline we simulate. Every step is a single global setting. There are no per-letter adjustments.

4 Identifying the font

The first question is which typeface the artist used. We restricted ourselves to fonts that existed in 1995–96 and that a Nintendo artist could plausibly have had:

- Macintosh system and LaserWriter fonts: Apple System 7.5.3 Palatino and Times, New York;
- Adobe PostScript fonts: IRIX 5.3 Type 1, plus X11 Adobe bitmap fonts;
- Microsoft's 1996 core fonts;
- classic Mac bitmap fonts with QuickDraw styles;
- the game's own dialog font, extracted from the ROM;
- free metric-compatible clones of the 35 PostScript fonts (URW base35).

Modern fonts were excluded.

Tournament (≈ 530 fonts and variants). A flexible model first screened 393 font files, then refined the best 157. It was flexible in that it allowed scale, aspect, blur, gamma and small per-letter shifts. The best member of each family is shown in Fig. 4.

- **Palatino wins clearly** (554), ahead of Times (638), New Century Schoolbook (694) and Georgia (745).
- The SM64 dialog font (812), Helvetica/Arial (890) and all others are far behind.
- Within Palatino, the 1980s/1990s design (URW P052, Apple 1996, Adobe 1990) fits better than the revised Palatino Linotype of 1998/99, which scores 633.

Identical rigid process search for every family. The flexible model could favour fonts with odd spacing, so we repeated the comparison with the *rigid* designer model. Every font received exactly the same exhaustive search over the following settings:

- 4 canvas shapes and 10 sizes;
- tracking, kerning and two advance modes;
- 4 word spaces, 2 filters, 2 blurs and 4 gammas;
- all positions.

Figure 5 and Table 1 show the result. The best font is Palatino Bold Italic (Adobe Type 1, 1990) (SSE 733.3), and the best non-Palatino font, N. C. Schoolbook Bold Italic (URW), is 108 worse. Figure 6 shows what each family's best recipe actually prints: only the Palatino italics reproduce the narrow, slanted, high-contrast letters of the real plaque.

The in-game font. Many players assume the plaque uses *Super Mario 64*'s own dialog font. We tested this directly.

- **Setup.** The 71 glyphs and the game's letter widths are extracted from the ROM (US, segment 2, IA4). The line is typed in the game font and scaled up, either as pixel art (nearest neighbour) or with Photoshop's default bicubic. It then goes through exactly the same canvas, resize, blur, gamma and position search as every other font.
- **Result.** Its best "Eternal Star" error is 1288 on the comparison grid and 1279 with the full grid (7.8 times the noise floor), against 757–759 for Palatino Bold Italic. That ranks it 31 of 33 fonts.
- **Flexible tournament.** The game font also loses there (812 vs. 554).
- **Downstream decoding.** With the game font, a dictionary search for the second word does not return "Star" at the top: it was rank 86 in our earlier rounds.
- **Why.** The dialog font is a small bitmap font with one-pixel strokes, no serifs and no stroke contrast. The plaque's letters have the serifs and the thick–thin contrast of an old-style italic.
- **Verdict.** The plaque was *not* set in the game's font; it was typeset with a desktop font.

Consistency checks.

1. **Two independent Palatinos agree.** The free URW clone (P052) and the genuine Apple 1996 TrueType Palatino independently converge on the *same* process recipe (Section 5).

#	font	family	SSE	Δ to best
1	Palatino Bold Italic (Adobe Type 1, 1990)	Palatino	733.3	+0.0
2	Palatino Bold Italic (URW P052)	Palatino	756.9	+23.6
3	Palatino Bold Italic (Apple 1996)	Palatino	759.2	+25.9
4	Palatino Italic (URW P052)	Palatino	801.6	+68.3
5	Palatino Italic (Adobe Type 1)	Palatino	802.9	+69.6
6	Palatino Italic (Apple 1996)	Palatino	809.1	+75.9
7	N. C. Schoolbook Bold Italic (URW)	New Century Schoolbook	841.2	+108.0
8	Times Bold Italic (Apple 1996)	Times	847.9	+114.6
9	N. C. Schoolbook Bold Italic (Adobe Type 1)	New Century Schoolbook	882.6	+149.3
10	Times Bold Italic (URW Nimbus)	Times	891.8	+158.5
11	Times Bold Italic (Adobe Type 1)	Times	913.4	+180.1
12	Times Bold Italic (Adobe, IRIX 5.3)	Times	913.4	+180.1
13	Georgia Bold Italic (MS 1996)	Georgia	956.1	+222.8
14	Adobe Garamond Bold Italic (Type 1)	Garamond	963.2	+230.0
15	Zapf Chancery (URW Z003)	Zapf Chancery	971.6	+238.4
16	N. C. Schoolbook Italic (URW)	New Century Schoolbook	984.5	+251.3
17	Times Italic (Apple 1996)	Times	992.8	+259.5
18	Palatino Bold (URW P052)	Palatino	997.0	+263.7
19	Zapf Chancery (Adobe Type 1)	Zapf Chancery	1007.4	+274.1
20	Times New Roman Bold Italic (MS 1996)	Times	1007.5	+274.2
21	Georgia Italic (MS 1996)	Georgia	1030.4	+297.1
22	Palatino Bold (Apple 1996)	Palatino	1033.8	+300.6
23	Palatino Roman (URW P052)	Palatino	1056.3	+323.1
24	Palatino Roman (Apple 1996)	Palatino	1074.3	+341.1
25	Bookman Light Italic (URW)	Bookman	1136.5	+403.2
26	Bookman Demi Italic (Adobe Type 1)	Bookman	1222.0	+488.8
27	Helvetica Bold Italic (URW Nimbus)	Helvetica / Arial	1229.8	+496.5
28	Bookman Demi Italic (URW)	Bookman	1232.4	+499.1
29	New York (Apple 1996)	New York	1255.2	+521.9
30	Helvetica Bold Obl. (Adobe, IRIX 5.3)	Helvetica / Arial	1263.6	+530.3
31	SM64 in-game dialog font (from ROM)	SM64 dialog font	1288.0	+554.7
32	Avant Garde Demi Obl. (URW)	Avant Garde	1486.7	+753.4
33	Courier Bold Italic (URW)	Courier	2109.6	+1376.3

Table 1: Font ranking under the identical rigid process search (SSE of the best recipe for “Eternal Star”).

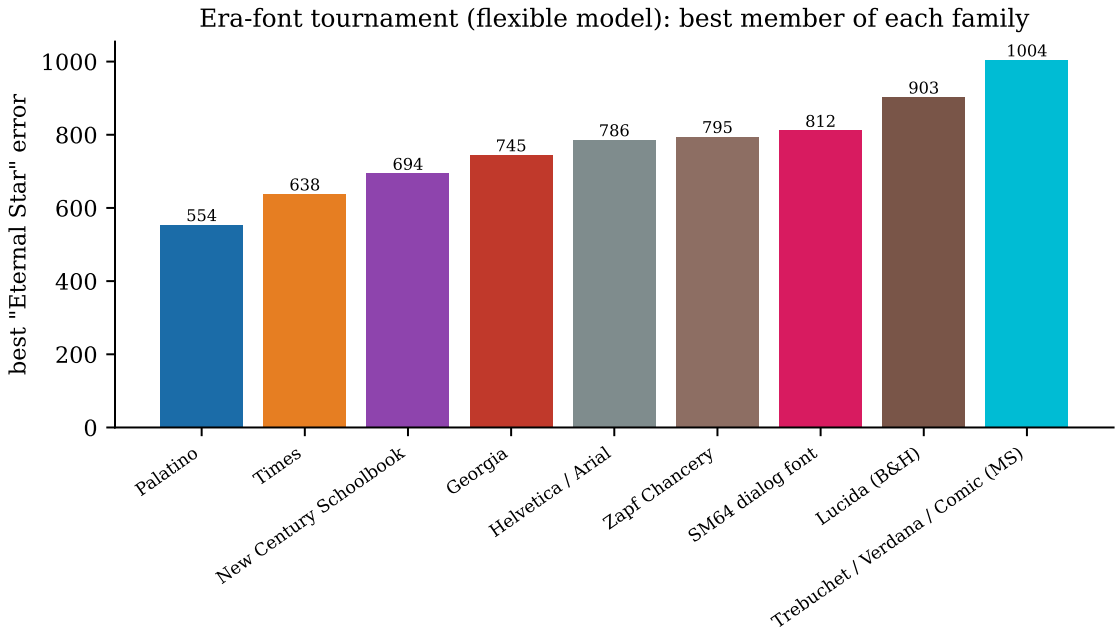


Figure 4: Era-font tournament, flexible model: the best "Eternal Star" error of each font family.

2. **Only Palatino decodes cleanly.** When the second word of line 1 or the words of line 2 are searched over a dictionary, Palatino returns "Star" and "hope" at rank 1. Times-family fonts return junk strings at the top, with "star" at rank 14–91.
3. **Weight and style.** Bold Italic and Italic are both close; upright Roman and Bold are clearly worse. The plaque is italic.

Verdict. The plaque is typeset in **Palatino Italic, most likely Bold Italic**, the classic Hermann Zapf design in its 1990s Mac/PostScript form. Among all Palatino digitisations, Adobe's original 1990 Type 1 file fits best; Section 10 identifies it as the artist's most likely font file.

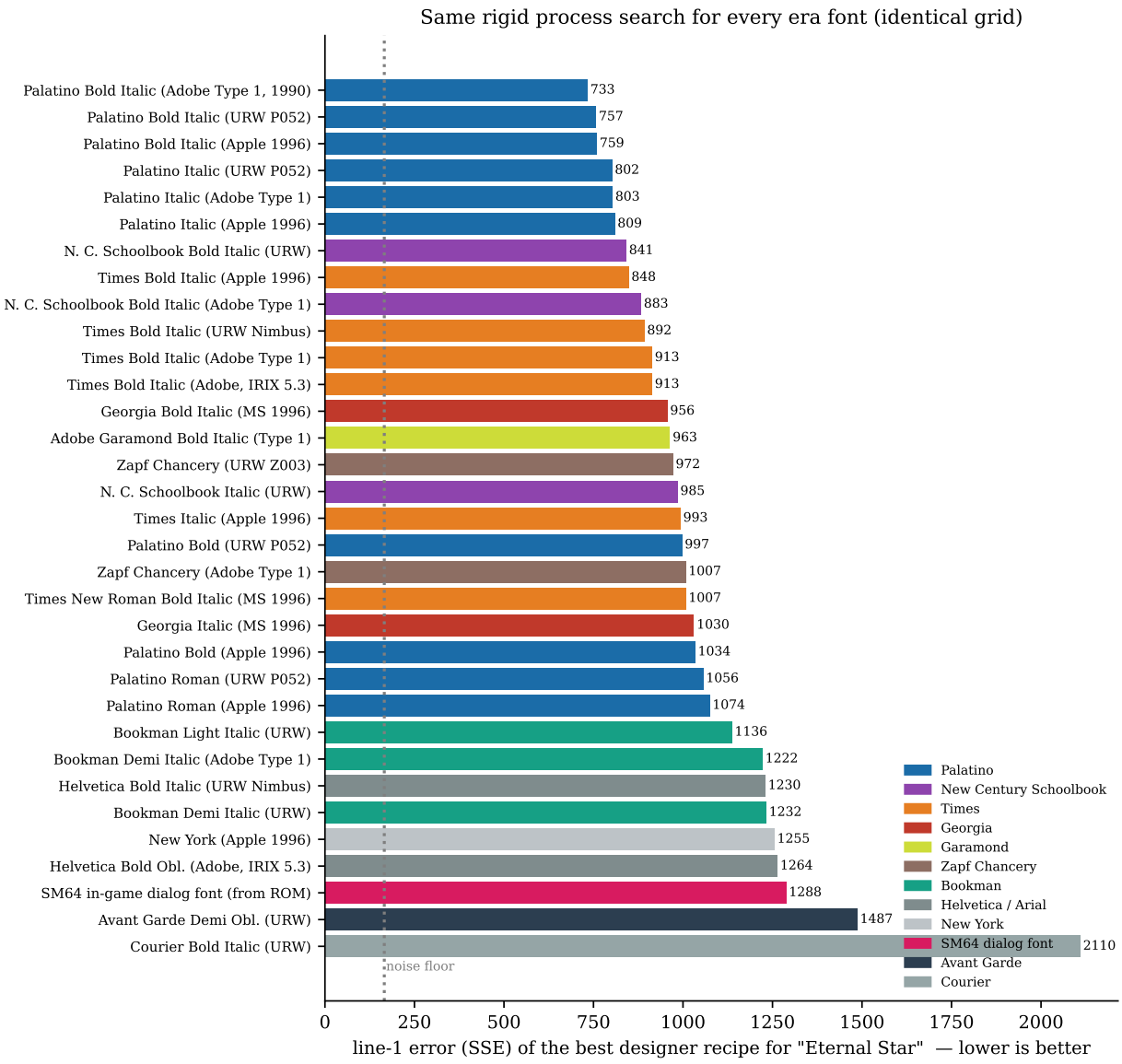


Figure 5: Same rigid process search for 33 era fonts: the best “Eternal Star” error per font, coloured by family.

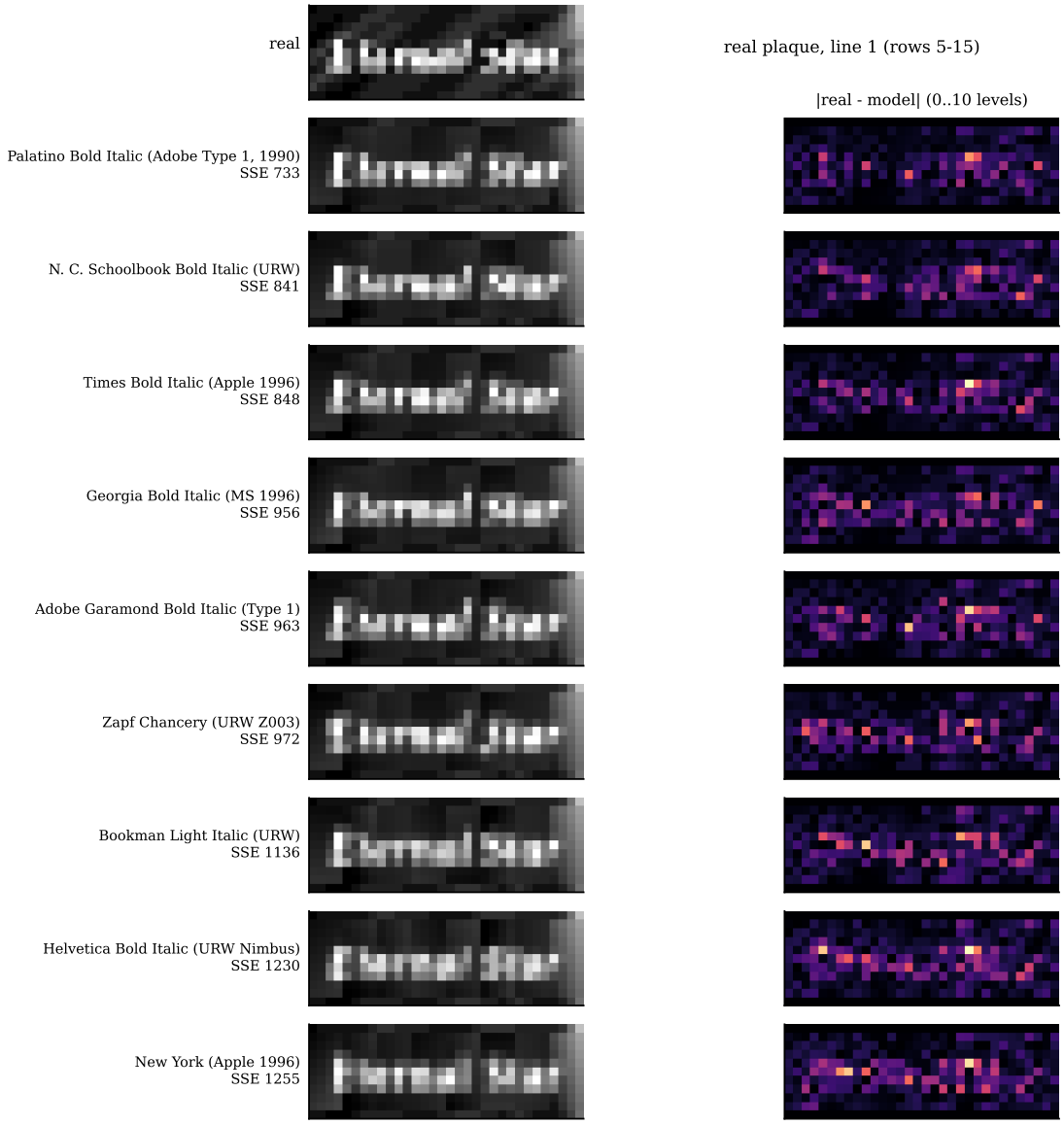


Figure 6: Line 1 as printed by the best recipe of each font family (left) and the absolute difference to the real plaque (right; 0–10 grey levels).

5 Reconstructing the process

Starting from a plain square canvas, we added one designer-plausible global step at a time (Fig. 7):

1. **Non-square artwork.** The art is drawn wider than tall and squeezed into the square texture. The quad itself is about 1.35:1.
2. **Word space, light blur and gamma.** The fit chooses $\gamma = 1.8$, the classic Macintosh display gamma.
3. **Whole-pixel advances.** Mac text without fractional widths places every letter on whole pixels.

The final recipe (Table 2) reaches SSE 757 (URW) and 759 (Apple), 4.6 times the noise floor. Both fonts choose the *same* canvas, size, tracking, filter, word space, blur and gamma. Adobe's own 1990 Type 1 file (first column; Section 10) prefers slightly larger artwork (192×160, 26 px) and fits better than both. The last two rows repeat the comparison against the extracted plain stone of Section 12: the order is unchanged.

What is still missing? A flexible model that may shift each letter horizontally reaches 548. With the per-letter shifts switched off, the same flexible model is *worse* than our rigid recipe (Fig. 8). The remaining gap is therefore pure letter spacing:

- "r n a l" sit about 0.3 px further right than these fonts place them;
- "Star" sits about 0.35 px to the left, which a narrower word space explains.

Eleven extra parameters on pure noise would buy about 6 SSE, not the 200 observed. The true 1996 spacing therefore genuinely differs slightly, through hinting by an unknown rasteriser or hand kerning. This affects neither the font family nor the words.

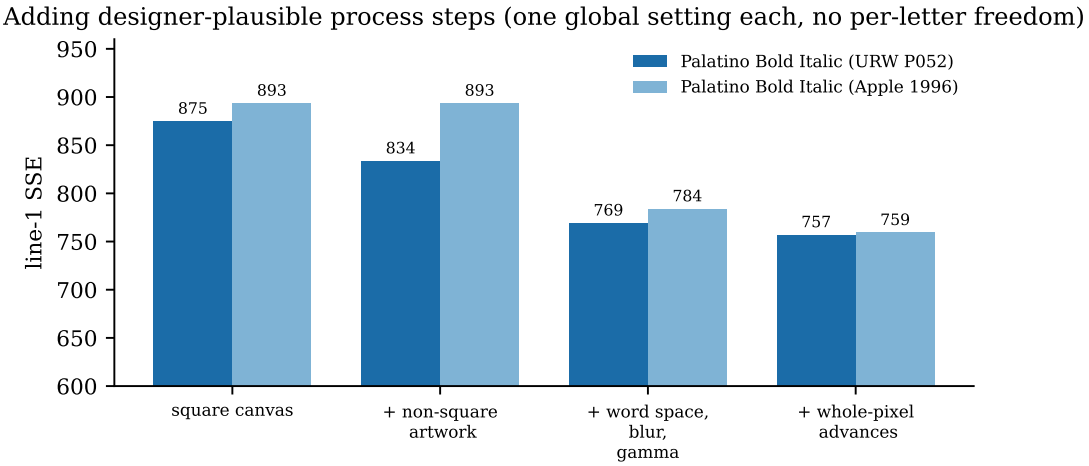


Figure 7: Each added process step is one global setting, chosen from a short list of options. The error drops from 875 to 757.

	Palatino Bold Italic (Adobe Type 1, 1990)	Palatino Bold Italic (URW P052)	Palatino Bold Italic (Apple
artwork canvas	192×160 px	160×128 px	160×128 px
font size	26 px	23 px	23 px
tracking	+2 px	+1 px	+1 px
pair kerning	on	off	on
advances	whole pixels per glyph	whole pixels per glyph	whole pixels per glyph
word space	+0.0 texture px	-0.5 texture px	-0.5 texture px
resize filter	bilinear	bilinear	bilinear
blur after resize	σ = 0.30 px	σ = 0.45 px	σ = 0.45 px
gamma	1.4	1.8	1.8
line-1 SSE (stone planes)	731.3	756.9	759.2
SSE / noise floor	4.41	4.57	4.58
same recipe, fixed stone	788.7	823.8	811.4
best recipe, fixed stone	788.7	814.0	811.4

Table 2: The reconstructed designer recipe for line 1: Adobe's 1990 file and the two independent Palatinos.

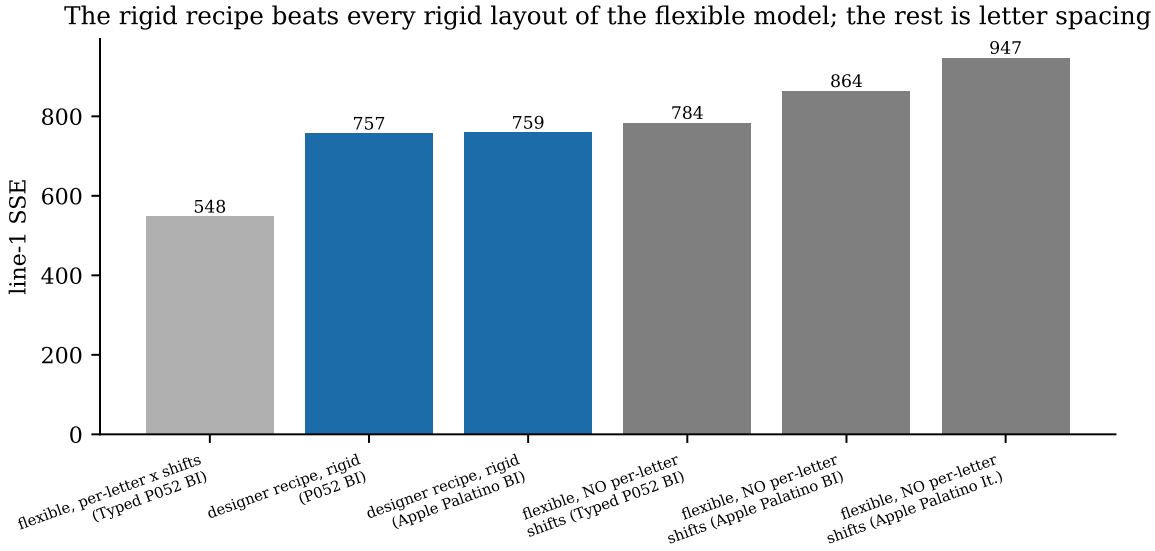


Figure 8: Ablation. The rigid recipe (blue) beats every rigid layout of the flexible model (grey); only free per-letter shifts do better.

6 Line 1: “Eternal Star”

“Eternal”. The first word is robust across every font, process and test. Its letters E, t and l are individually confirmed by local fits.

The second word, with every setting frozen. We froze the complete recipe (font, canvas, size, tracking, filter, blur, gamma and position) and rendered “Eternal ___” for each of 27,122 dictionary words. Only the word space (4 values) and the ink level remained free.

- “Star” is **rank 1 in both fonts** (Fig. 9).
- The runners-up (Stat, Sent, Seat) share its shape: S, a tall letter, two small letters.

Fairness. The recipe was tuned on “Star”, so we re-ran the *entire* recipe search separately for each rival (Fig. 10).

- “Star”/“star” still beat Stat, Sent, Slot and Seat.
- On line 1 alone, upper- and lower-case “s” are tied: 757 vs. 755 (URW) and 759 vs. 765 (Apple).
- **Joint two-line test.** One process must explain both lines. For each spelling we took its 100 best line-1 recipes and scored line 2 (“We hope so”) under each of them, then compared the best totals.
  - Genuine Apple font: “Eternal Star” 1132.5 vs. “Eternal star” 1176.3, so the capital S is ahead.
  - URW clone: 1101.9 vs. 1072.2, so the lower-case s is ahead.

  The two fonts disagree, so **the case of the s remains undecided**. We write “Eternal Star”, but the pixels do not settle it.

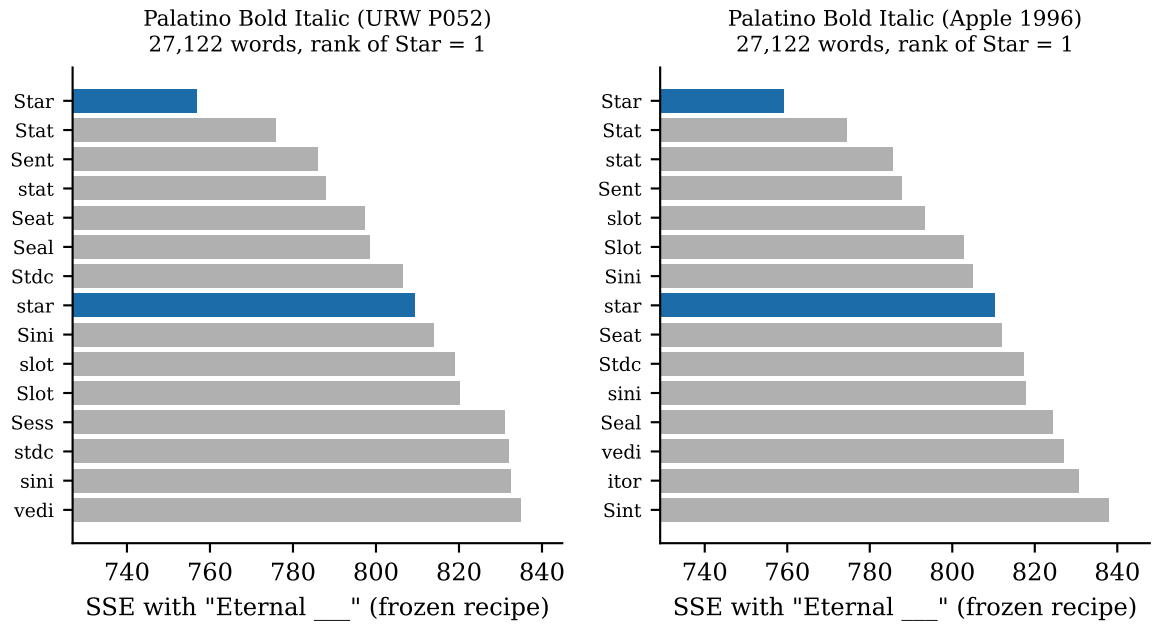


Figure 9: The second word of line 1 under the frozen recipe: top 15 of 27,122 words.

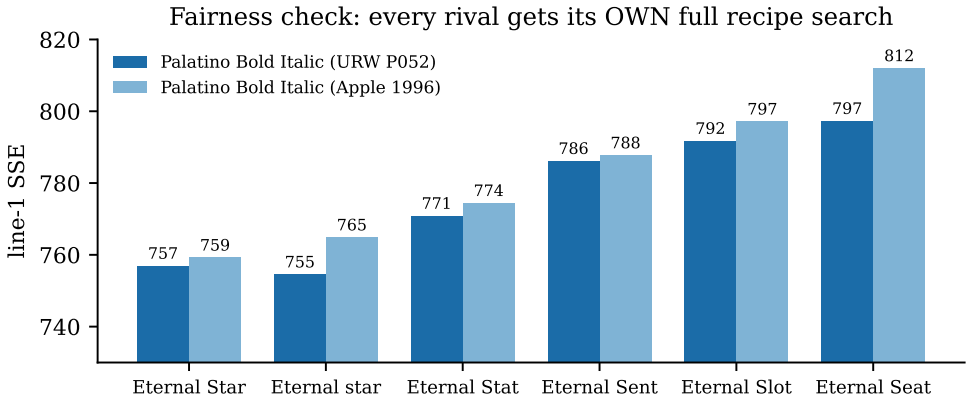


Figure 10: Every rival second word gets its own complete recipe search.

7 Line 2: "We hope so"

Locked process. We assume the designer set line 2 in the same document. It therefore shares the face, canvas, tracking, advance mode, filter, blur, gamma and *ink level* of line 1. This removes the contrast freedom that let nonsense win in earlier attempts. Only the size, the baseline, the start position and the word space are free.

Fast exact-enough search. Every glyph at every canvas position is pre-rendered through the process as a small "tile", and a string is a sum of tiles. A K-best dynamic programme then finds the best *word sequences*: any number of words, any start position, over a 17.7k-word English vocabulary. It runs for every size (13–20 px), baseline and word space. The top 3,000 phrases are re-rendered exactly.

What comes out unprompted.

- **Size and spacing:** 17 px, with exactly line 1's narrower word space.
- **Centring:** the line sits under line 1 (ink centre column 15.0 vs. 15.5), consistent with centred text.
- **Words:** 45 of the top 60 sequences contain the word "hope", and almost all begin with a W-word (Fig. 11).

Exhaustive slot tests (Fig. 12).

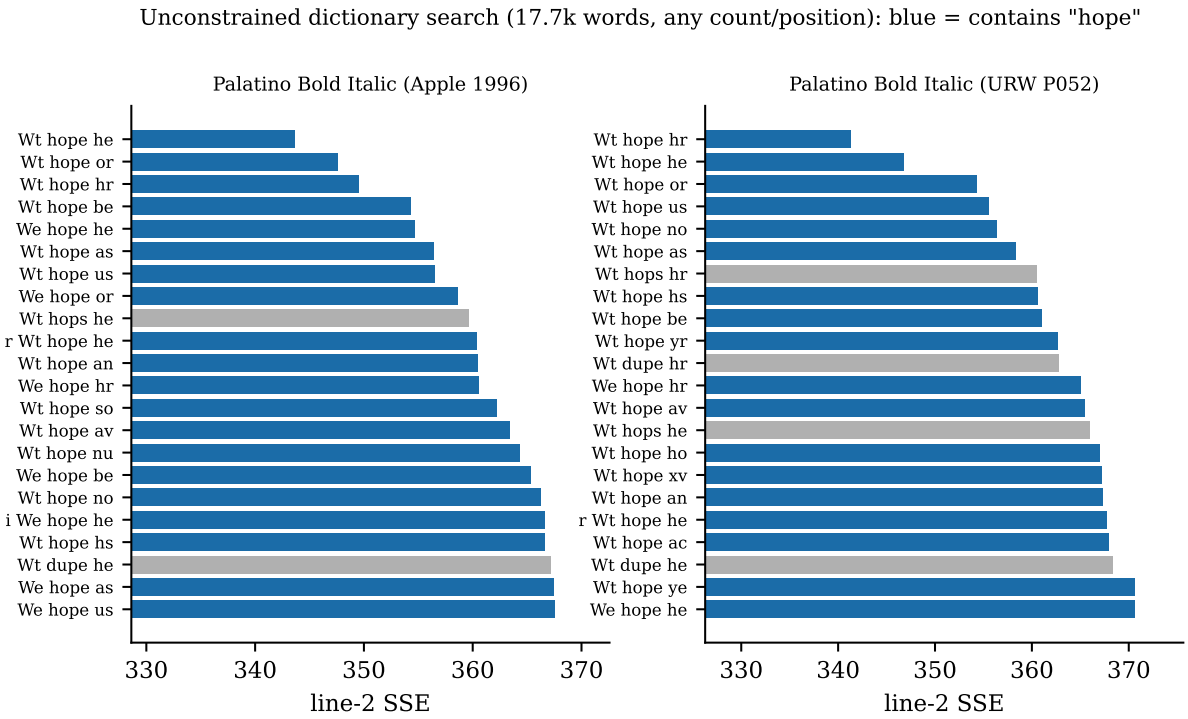


Figure 11: Unconstrained dictionary search for line 2. Without being told anything, "hope" appears in nearly every top sequence (blue).

- **Middle word.** In "We ___ so", "hope" is **rank 1 of 12,556 words in both fonts**, 16–19 SSE ahead of "hops"; everything else is more than 24 behind.
- **First word.** Among all 157k strings of one to three letters, the pixels want a capital W followed by one small letter. "We" is the best English word, ahead of "Me" and "He".
- **Ending.** The ending is two x-height letters. "he, or, us, as, be, no, an, so" lie within about 20 SSE of each other.

The ending under line 1's typesetting. Table 3 lists the main variants. The simulations in Section 9 show that two-letter endings within about 20 SSE cannot be separated by the pixels: a planted "We hope so" was decoded as "We hope an". Of the pixel-compatible endings, "so" is the only one that forms a complete sentence. Once line 2's own typesetting is calibrated (Section 8), "so" also becomes the best-fitting real ending.

Several alternatives are clearly worse:

- a Zapf Dingbats star, punctuation-only endings and Japanese-style "~" endings;
- "We hope you";
- Title Case "We Hope So";
- no ending at all, which scores 682.

Descenders. Row 20 lies just below the line-2 baseline. It is significantly brighter than the stone below it in *one* place only: column 16 (+5.0 levels; clean-stone s.d. 1.14), which is exactly where the "p" of "hope" falls (Fig. 14). A reading with two descenders, such as "My hope is", predicts a second descender under the "y" that the texture does not show.

Close-up: candidates for "We" and for "so"

These are the two least certain words, so we isolated them. In each test the rest of the line is held fixed: either "___ hope so" or "We hope ___". Each candidate is rendered as a whole line with the locked recipe, but scored only on the stone blocks that contain the word in question:

- first word: texture columns 0–11;
- ending: texture columns 20–31.

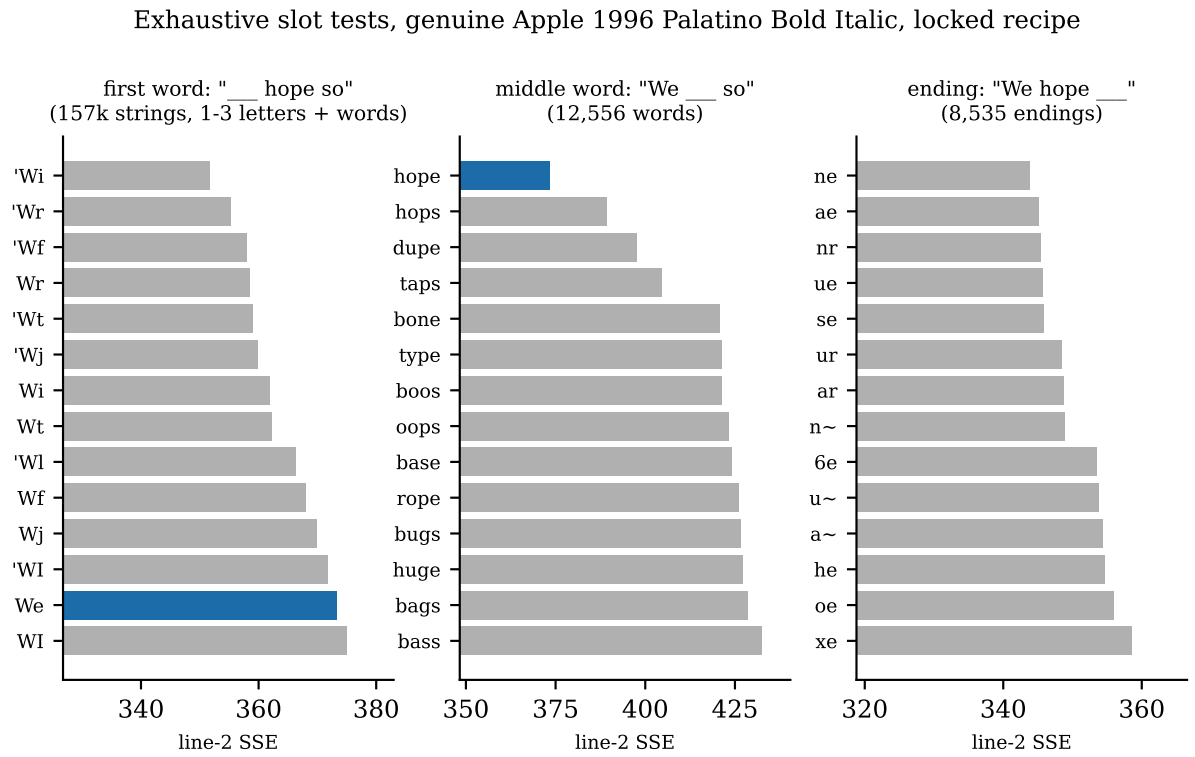


Figure 12: Slot tests with the genuine Apple 1996 Palatino Bold Italic: first word, middle word and ending.

Because each region is scored on its own pixels, the comparison is not diluted by the rest of the line. The best non-word strings found by the exhaustive search are shown for reference (Figs. 15 to 17).

First word.

- **What the pixels want.** A wide capital with a double-bowl or double-V top, followed by small letters. The non-words "Wi", "Wr" and "Wt" sit at the very top in both fonts.
- **Best real words.** "We'll", "We" and "By" are a tight group on their own region: "We" scores 187 (Apple) and 171 (URW), and "By" 176 (Apple).
- **Clearly worse in both fonts:** "Me", "He", "My", "No" and "Be".
- **Whole-line test.** A local fit is not enough, because the first word's width fixes where "hope so" must sit. On the whole line, "We'll" and "By" push "hope so" out of place and fail (Section 8); only "We" remains.

Ending.

- **Shape.** The ending is two x-height letters, with a bright stroke at the start and a rounded second letter. The best non-words ("ne", "nr", "ae") show exactly this.
- **Ranking of real endings.** In both fonts: "he" (113 / 114), then "or", "us", "as", "be", "no" (the order varies slightly between fonts), then "so" (132 / 134).
- **Clearly worse:** "to", "for" and "is". Their ascenders and the long "for" add ink where the texture has none.
- **What can be resolved.** The spread from "he" to "so" is about 20 local SSE. The simulations (Section 9) show that differences of this size between two-letter endings cannot be resolved: a planted "We hope so" came back as "We hope an".

Variants. Table 3 and Figs. 18 and 19 compare whole-line variants.

- The four best-fitting lines are "We hope he", "…or", "…us" and "…no". None of them is a complete sentence.
- **The best-fitting *complete* sentence is "We hope so"**, in both fonts. It is followed by "We hope so." and "We hope on".
- "We hope to", "We hope is", "We hope for", "My hope is", "He hopes so" and "I hope so" are clearly worse.

line 2	Apple	URW	mean	complete sentence?
“We hope he”	355	371	363	no
“We hope or”	359	378	368	no
“We hope us”	368	379	373	no
“We hope no”	377	380	379	no
“We hope so”	373	389	381	yes
“Me hope so”	392	385	389	no
“We hope so.”	386	392	389	yes
“We hope on”	382	408	395	yes
“We hope to”	397	416	406	no
“We hope so!”	411	407	409	yes
“We hope is”	404	421	412	no
“My hope so”	435	410	423	no
“He hope so”	441	429	435	no
“We hope for”	460	434	447	no
“No hope so”	461	464	463	yes
“My hope is”	481	459	470	yes
“We hope it”	467	477	472	no
“my hope is”	489	475	482	yes
“We hope all”	510	461	485	no
“We Hope So”	523	481	502	yes
“We hope one”	514	492	503	no
“We hope too”	532	484	508	no
“’Tis hope so”	554	471	512	no
“To hope is”	523	538	530	yes
“We hope for it”	550	524	537	yes
“We hope ~”	522	585	553	no
“We hope you”	564	587	576	no
“We hope to see”	647	628	638	yes
“We hope”	682	696	689	no
“We hope!”	716	720	718	no
“He hopes so”	742	743	742	yes
“I hope so”	1047	1024	1035	yes

Table 3: Exact line-2 errors (whole band) of the main variants in both fonts, and whether each forms a complete sentence. Line-2 noise floor 165.7; stone only (no text) 2172.8.

Other first words and other sentences. We also tried alternatives suggested by the first-word close-up, and sentences that would make good sense on such a plaque, including quirky Japanese-English phrasings a 1996 Japanese team might have written.

- **Seeded search.** The global dictionary search was seeded with the first words We, By, Be, Me, My, He (and variants), with everything after them free. A further 91,884-phrase set combined these first words with about 90 middle words and 40 endings, in sentence and Title case.
- **“hope” survives every first word.** The best line for each first word is: “We hope he” (355), “Me hope he” (374), “He hope he” (389), “My hope he” (417), “By hope or” (420) and “Be hope us” (459), all in Apple Palatino.
- **“We” gives the best fit by a wide margin.** “By” matches its own region well, but its width pushes “hope” out of place.
- **Other sentences are far outside the pixel-compatible window** (Table 4, Fig. 20): “Be with us” and its variants, “By your side”, “We love you”, “Be happy”, “We pray so” and “We hope you”. None of them reproduces the tall-first-letter, mid-word-descender shape of the middle word.



Figure 13: Line 2 band: real pixels, model and difference for several readings.

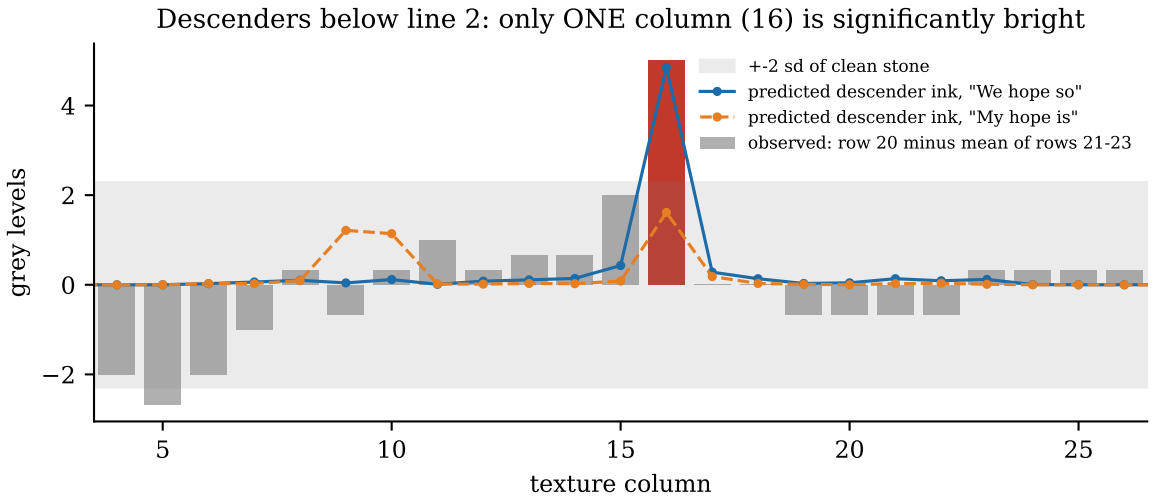


Figure 14: Descender evidence. Bars: brightness of row 20 relative to the stone below. Lines: descender ink predicted by "We hope so" and "My hope is".

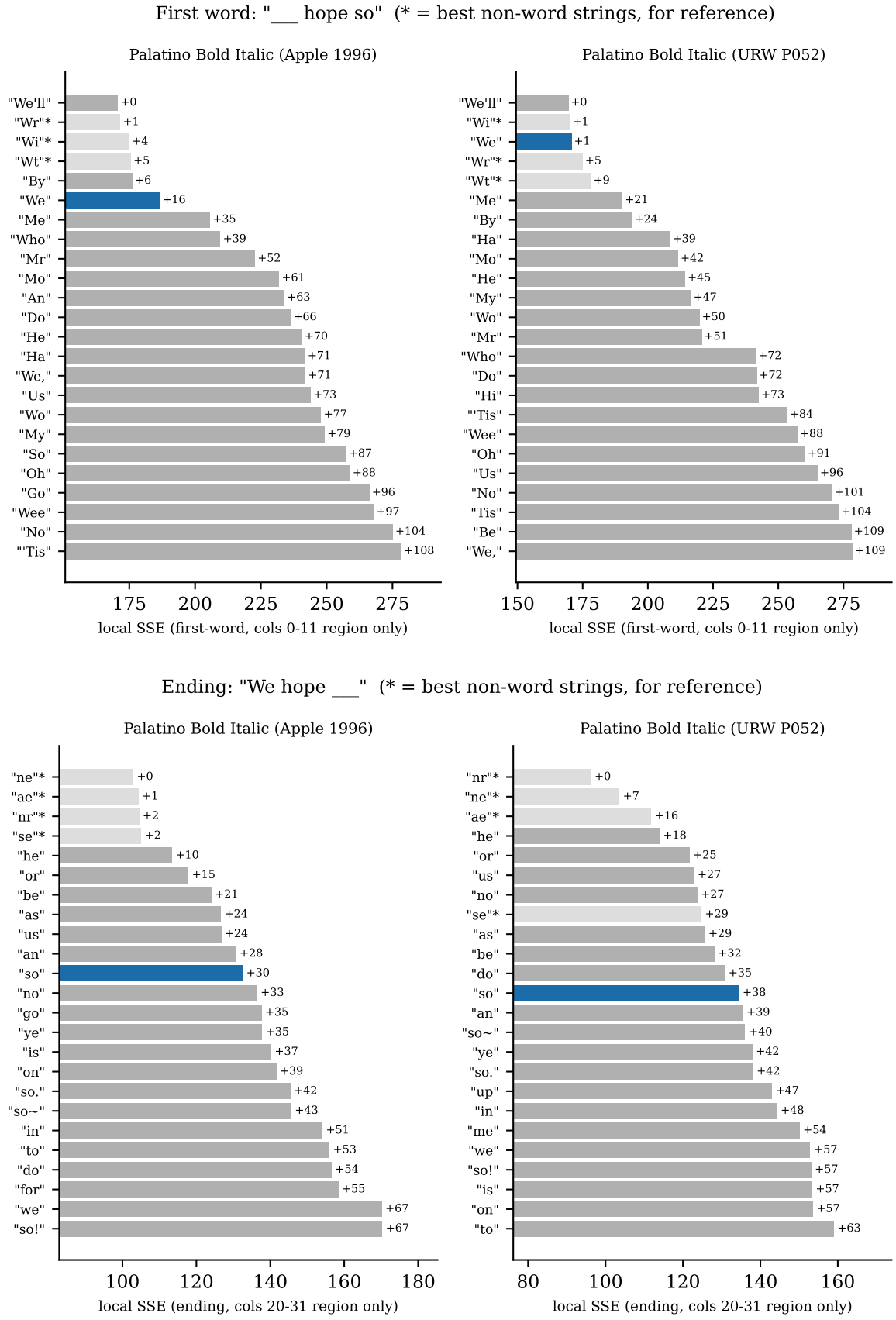


Figure 15: Local errors of first-word candidates (top) and ending candidates (bottom) in both fonts. Blue: our reading. Light grey with *: best non-word strings, for reference.

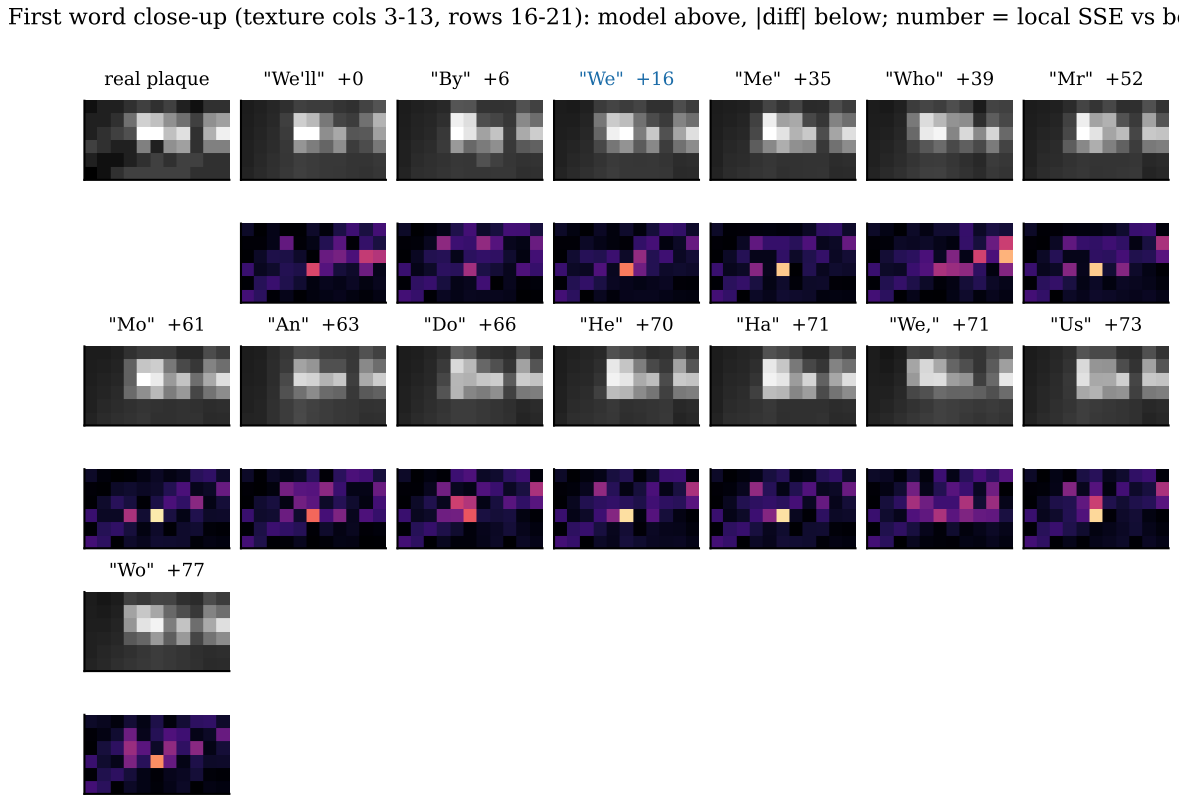


Figure 16: Close-up of the first word (Apple Palatino). For each candidate, the model of the zoomed region is shown above and the absolute difference to the real plaque below. The number is the local SSE relative to the best candidate.

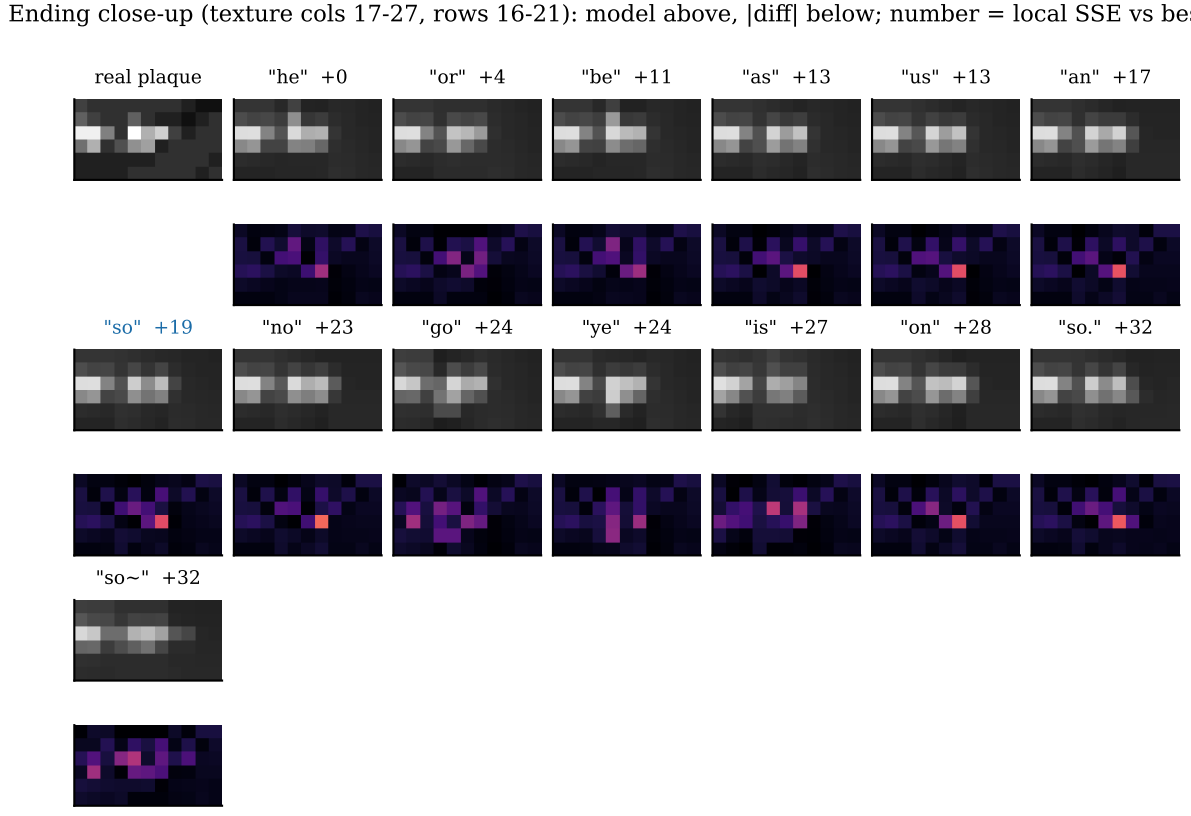


Figure 17: Close-up of the ending (Apple Palatino), laid out as in Fig. 16.

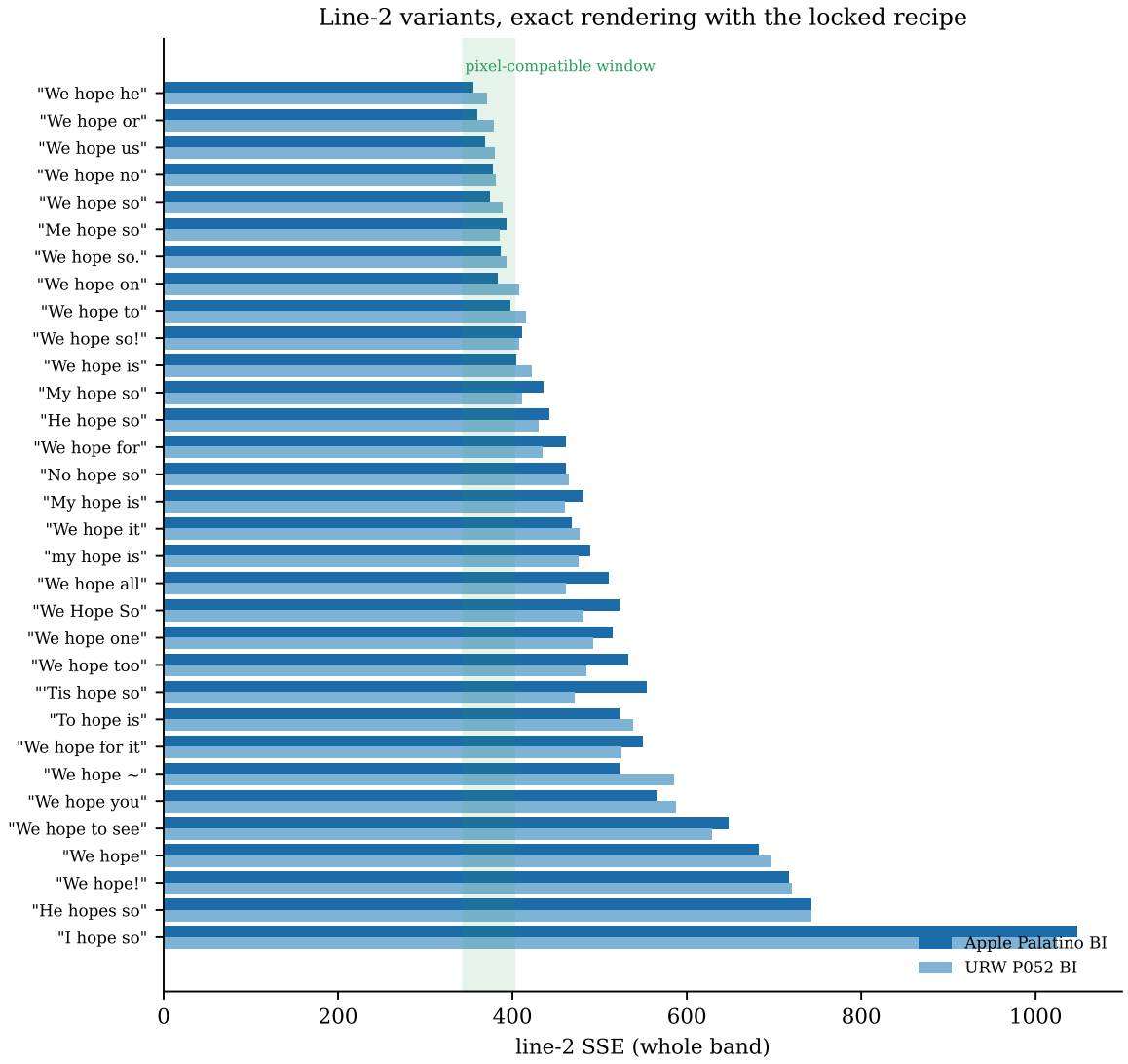


Figure 18: Whole-line variants in both fonts. The green band is the pixel-compatible window (Section 9).

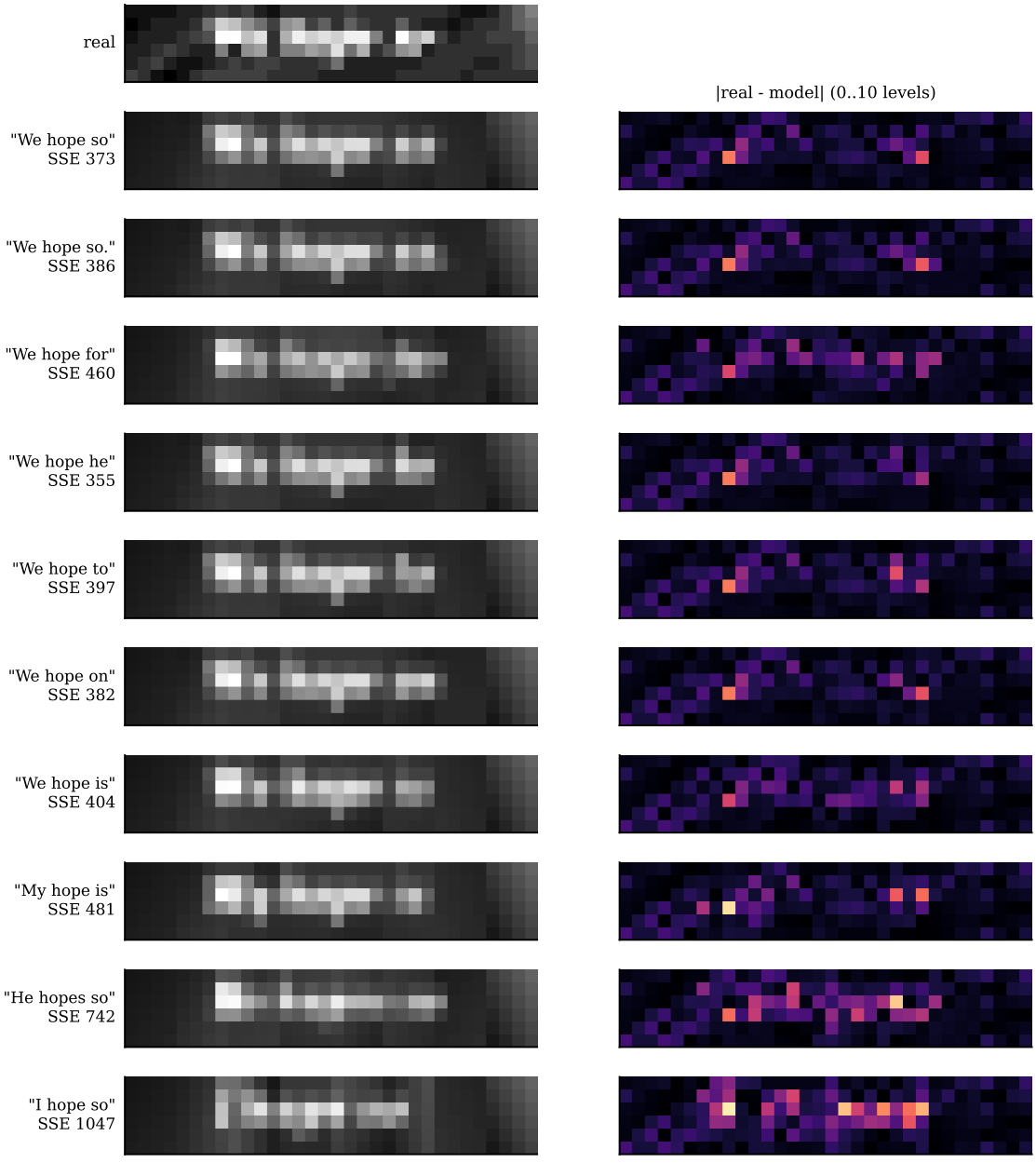


Figure 19: Line 2 for the main variants: model (left) and absolute difference to the real plaque (right).

line 2	Apple	URW	mean
“We hope he”	355	371	363
“We hope so”	373	396	385
“By hope so”	434	409	421
“Be hope so”	492	493	493
“We pray so”	550	537	543
“We hope you”	552	587	569
“We wish so”	675	555	615
“By my side”	653	654	654
“We hope it so”	695	662	679
“Be my star”	681	685	683
“Be with us”	730	666	698
“Be with me”	724	675	699
“Be with us~”	731	669	700
“Be happy so”	705	726	716
“Be with us!”	752	689	720
“Be happy”	730	720	725
“We love you”	708	751	729
“Be With Us”	780	736	758
“By your side”	760	766	763
“We are hope”	752	793	772
“Be my hope”	794	771	783
“We keep hope”	764	811	788
“Be here now”	837	796	817
“Be at peace”	873	798	835
“Be with you”	867	858	863
“Be our hope”	886	861	873
“Be here”	889	865	877
“Wish so”	906	926	916
“Be safe”	927	916	921
“We be ok”	933	924	928
“Hope so”	1000	947	973
“Be well”	1062	1053	1057
“Be ok”	1297	1264	1280

Table 4: Alternative sentences for line 2 (exact, locked recipe; lower is better).

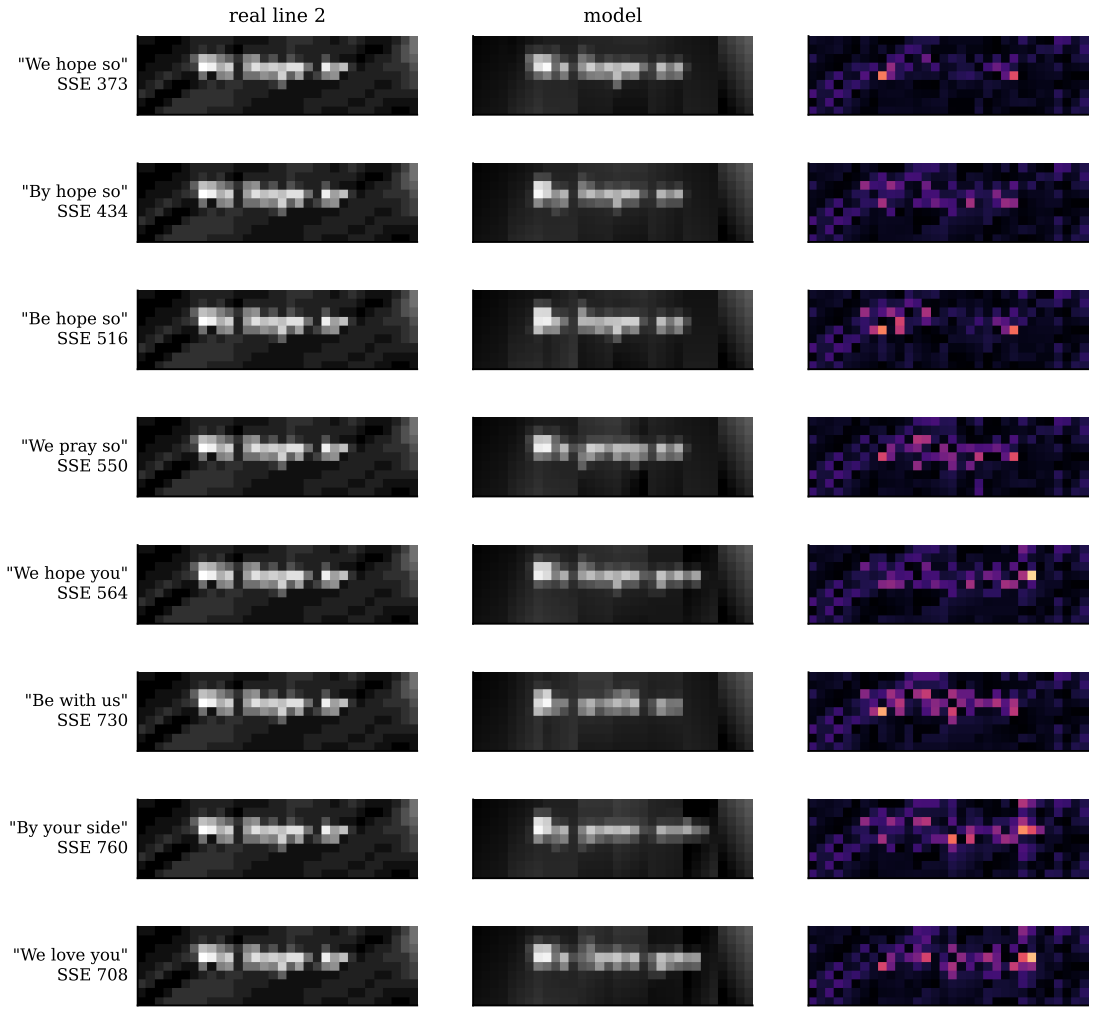


Figure 20: Line 2 for alternative sentences: model (middle) and absolute difference (right) next to the real plaque (left).

8 Working backwards from “hope”

For line 1 we assumed the text and solved for the process. We can do the same for line 2, using its most certain word.

What may differ between the lines. Canvas, resize filter, blur, gamma and ink colour are *shared*: both lines sit in one image, which is resized once. Each typed line may still have its own size, tracking, kerning, advance mode and even style. In Section 7 those were copied from line 1.

Fitting line 2’s typesetting on “hope” alone. We render only the word “hope” and score it only on its own pixels: the stone blocks covering texture columns 12–19. The search covers:

- four faces: Palatino Bold Italic and Italic, Apple and URW;
- sizes 13–21 px and tracking −1 to +3;
- kerning on/off and two advance modes;
- all positions.

Both Bold Italic faces fit equally well (44.5 and 45.2), at **18 px, tracking +0 (line 1: +1), whole-pixel advances and no kerning** (Fig. 21). The Italic (non-bold) faces fit worse.

Then judging “We” and “so” on their own pixels. This is a held-out test. The typesetting was calibrated on columns 12–19 only, and the first word and the ending are then scored on *different* pixels: columns 0–11 and 20–31 (Figs. 22 and 23).

- **First word (URW line 2).** “We” is the best real word (179). It is ahead of “Me” (202); “By” drops out of contention. Only the non-word “Wr” is lower (161).
- **First word (Apple line 2).** The W-forms lead again.
- **“We’ll”.** With the calibrated setting, “We’ll” matches its own region best of all, because its extra strokes resemble a faint mark near column 9. On the whole line, however, “We’ll hope so” scores 642 (URW), against 329 for “We hope so”. The wider word pushes “hope so” out of place. “We’d”, “Well”, “Will” and “We will” fail in the same way (544–1000). “By” shows the same effect, so a good local fit alone is not enough.
- **Ending. “so” becomes the best real ending in both faces**: 106 vs. “or” 118, “us” 120, “he” 124 (URW); 107 vs. “as”, “or” 115, “us” 118 (Apple). Only non-words such as “nr” or “se” are lower.
- **Whole line (URW).** “We hope so” becomes the best of all variants (338), ahead of “We hope or” (341), “We hope us” (343) and “We hope he” (346). It is also the best line-2 error of the whole study (Fig. 24).

The unconstrained search now finds the sentence on its own. We repeated the global dictionary search under the calibrated setting: 17.7k words, any number of words, any position, no hints (Fig. 25). The best strings are now “We hops So”, “Wt hope so” and **“We hope so”** (rank 3, 328.7; URW). With Apple Palatino the top is “We hops So” with “We hope So” at rank 3.

- Without any guidance, the search now produces *We* at the start and *so* at the end. Under line 1’s typesetting it had preferred “he”/“hr” endings.
- The only rival for the middle word is “hops”, which differs from “hope” by one small letter. That is at the limit of resolution; “hope” is the English choice.
- Upper- or lower-case “so” cannot be distinguished.

Why this matters. Calibrating the typesetting on one word made the other two words come out without being told. This is the same self-consistency that identified “Eternal Star”: one process and one font explain the whole line, and they predict “We” and “so” on pixels that were not used to fit them.

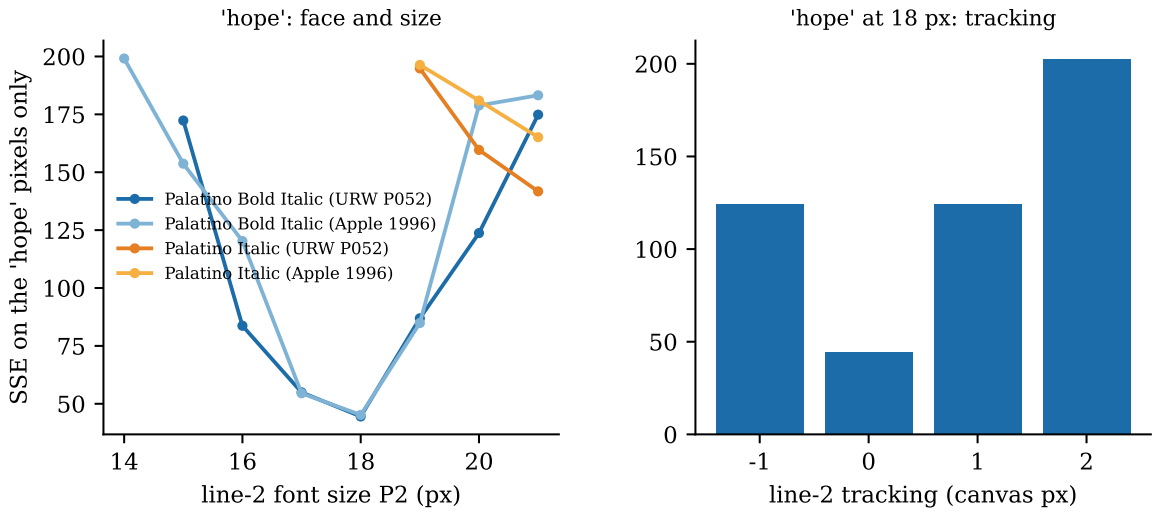


Figure 21: Line-2 typesetting fitted on the four letters of “hope” only. Left: error vs. font size for four faces. Right: error vs. tracking at the best size.

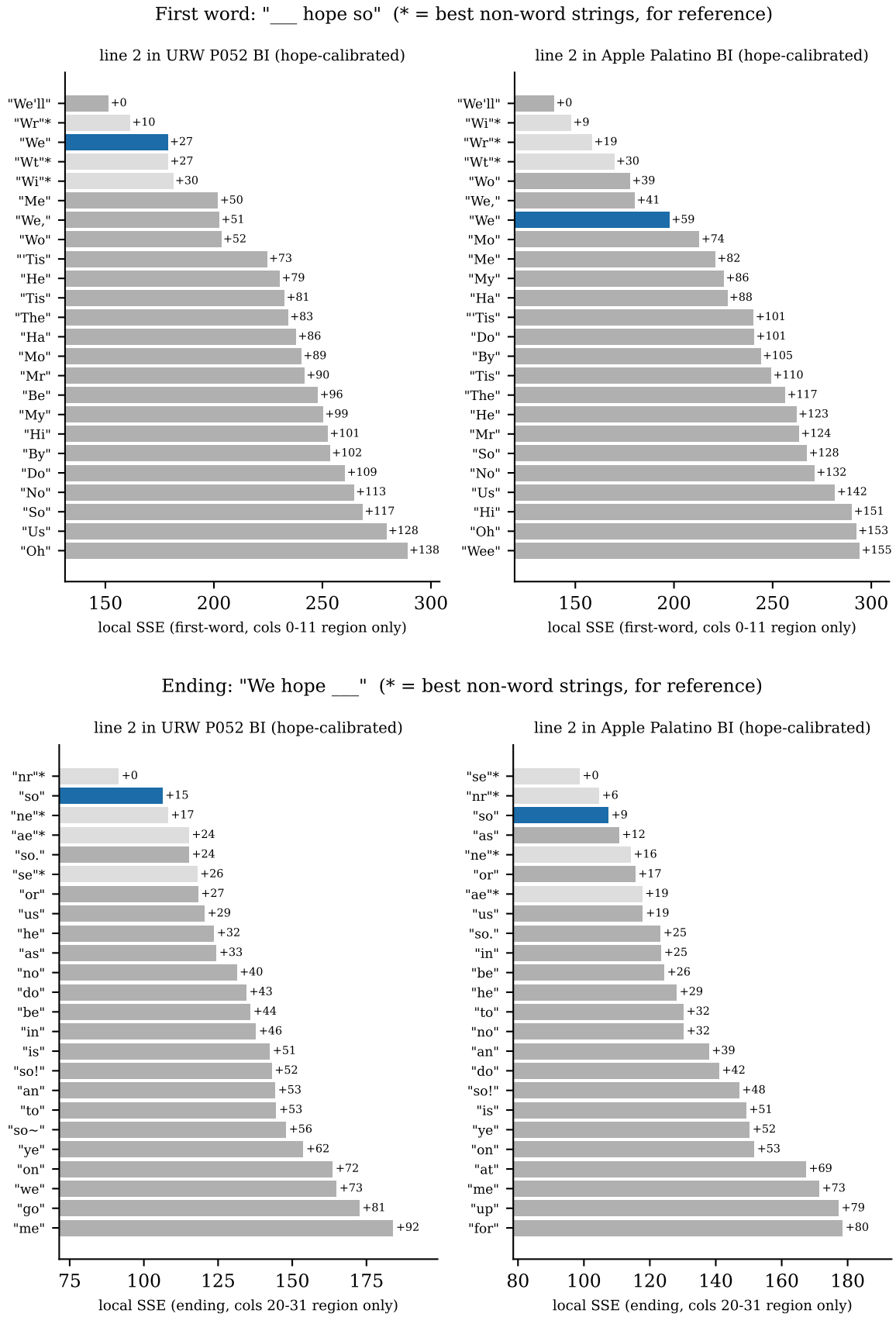
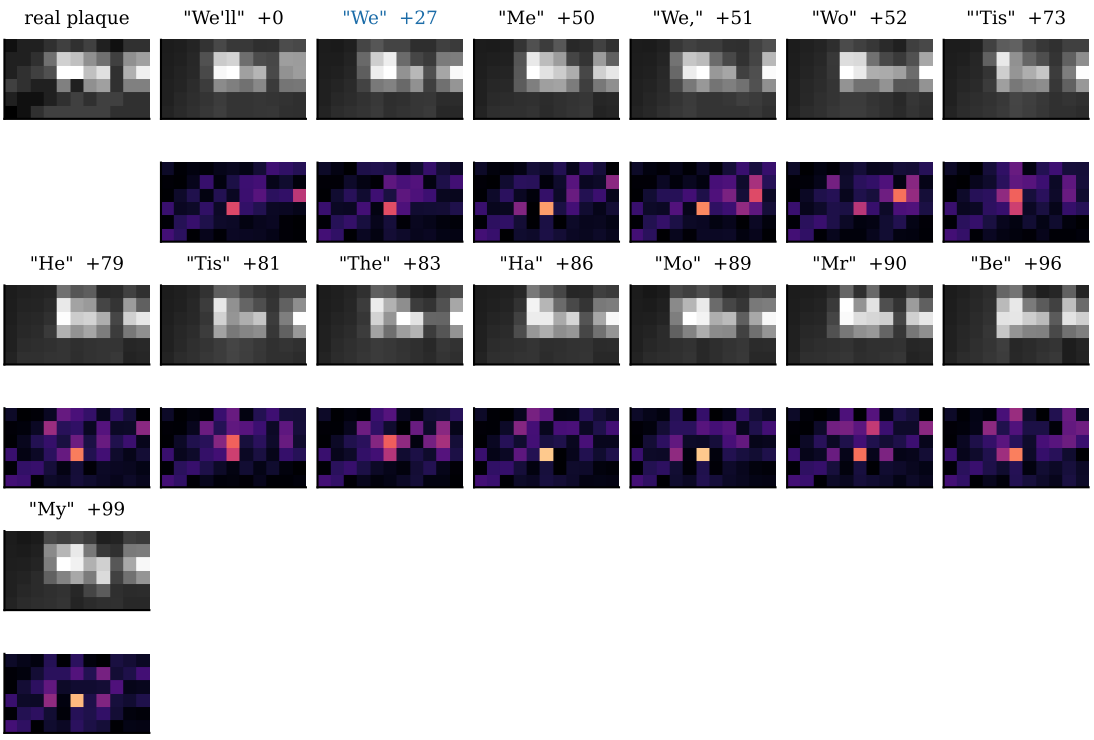


Figure 22: After calibrating line 2 on “hope”: local errors of first-word candidates (top) and ending candidates (bottom).



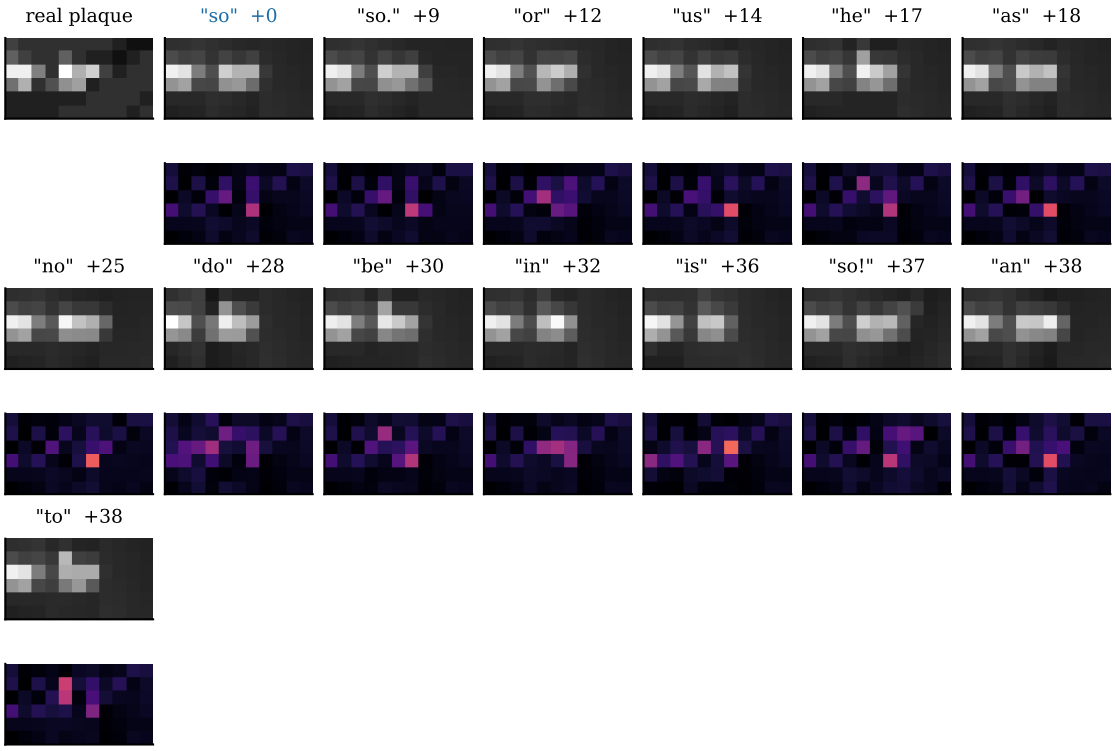


Figure 23: Close-ups after the hope calibration (URW line 2): first word (top) and ending (bottom). Model above, absolute difference below.

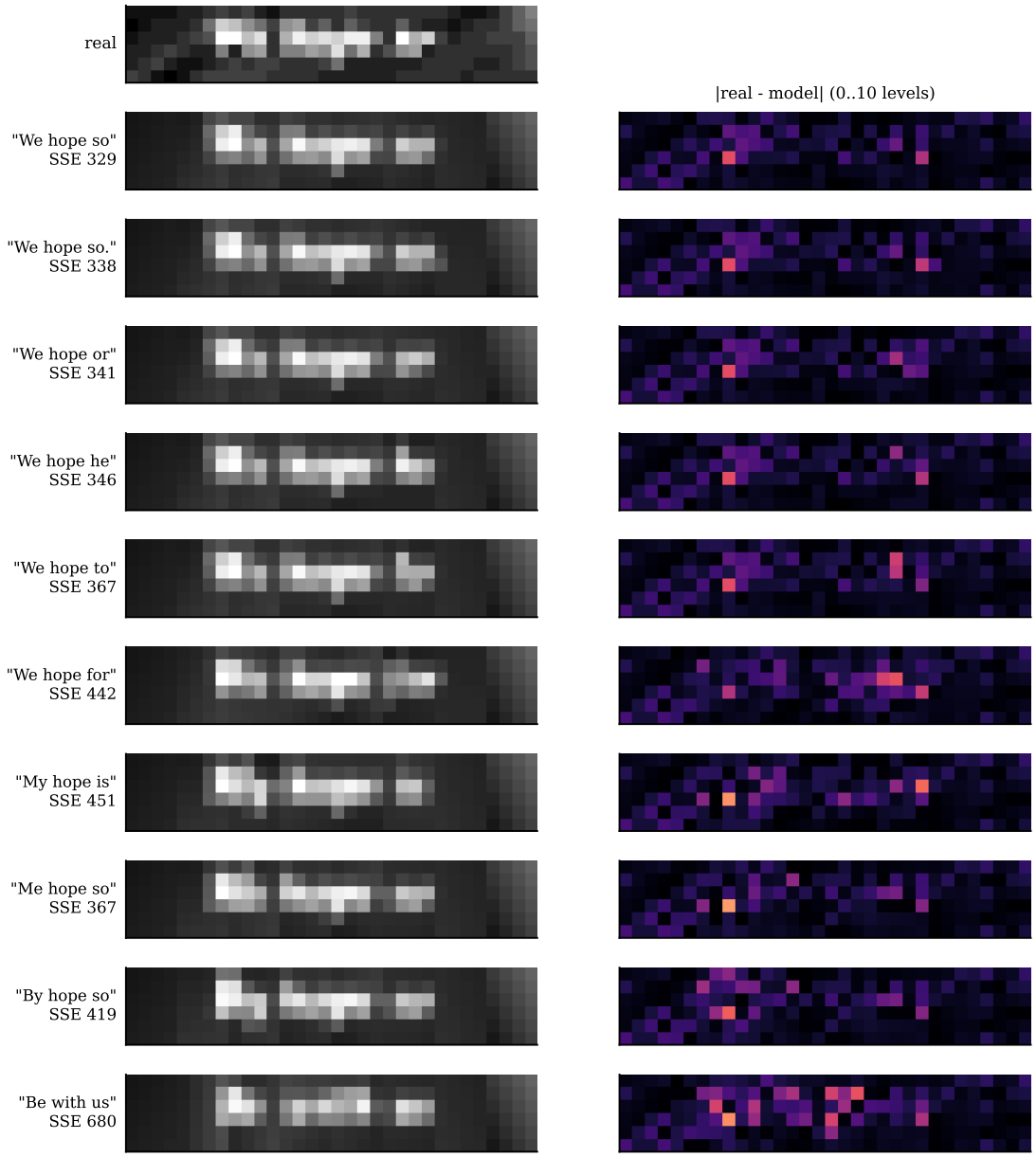


Figure 24: Whole-line variants under the hope-calibrated line-2 typesetting (URW): real plaque (top), model (left) and absolute difference (right).

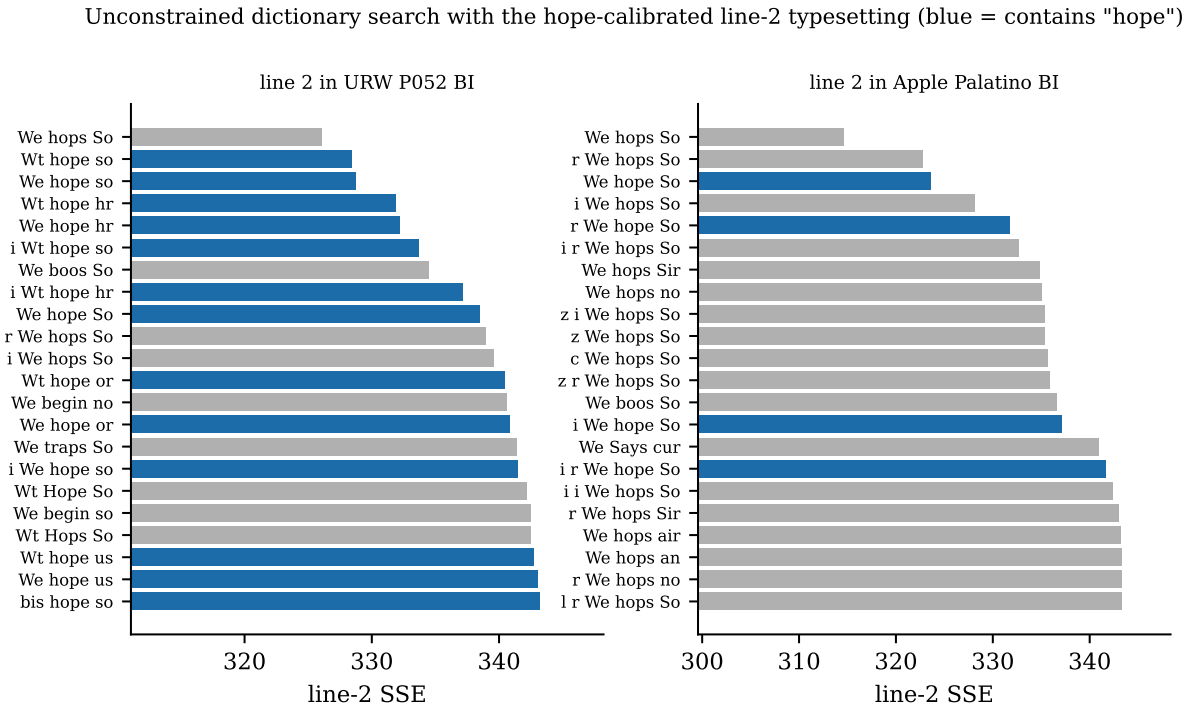


Figure 25: Unconstrained dictionary search under the hope-calibrated line-2 typesetting. "We …so/So" now emerges on its own.

9 Calibration: how much can the pixels tell?

We planted 24 known phrases into the real plaque. For each one we replaced the ink of the best reading by the phrase's ink and kept the real stone and the real fit residual. We then decoded each plaque blind with the same dictionary search (Fig. 27).

- **Exact recovery is rare.** The exact phrase came back first in 2/24 cases, and 32/68 words (47 %) were recovered. With a large vocabulary, some nonsense string often fits slightly *better* than the truth.
- **The truth is never far from the optimum.** The true phrase always scored within ≈ 58 SSE of the best string found. This defines a *pixel-compatible window*: readings more than about 58 above the optimum are effectively excluded, and readings inside it are distinguished by language.
- **Strong words survive.** "hope" (or "hops"), "Shine", "Thank you", "happy" and "light" were recovered well. Short function words and two-letter endings were not.
  Under line 1's typesetting, the optimum for the real plaque is about 341–344.
- **Inside the window, in both fonts:** "We hope so" (373 Apple, 396 URW).
- **Well outside:** "My hope is" (459–511), "No hope so" and "'Tis hope so" (466–474), and "lie here ~" (689–695).

Calibration under the hope-calibrated typesetting. We repeated the test with line 2's final typesetting from Section 8 (Fig. 26).

- 4/24 phrases were recovered exactly and 40 % of words were recovered.
- The truth always lay within 52 of the best string found.
- On the real plaque, "We hope so" lies only **2.6** above the global optimum (328.7 vs. 326.1 for the non-English "We hops So"). That is deep inside the window where, in simulation, the true text is found.

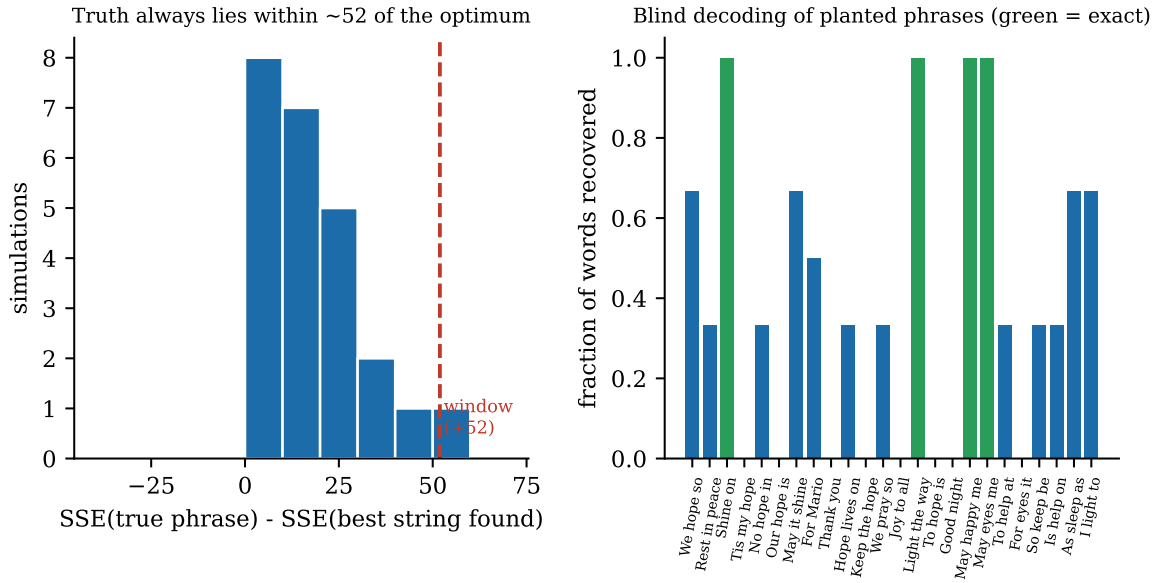


Figure 26: Calibration under the hope-calibrated line-2 typesetting.

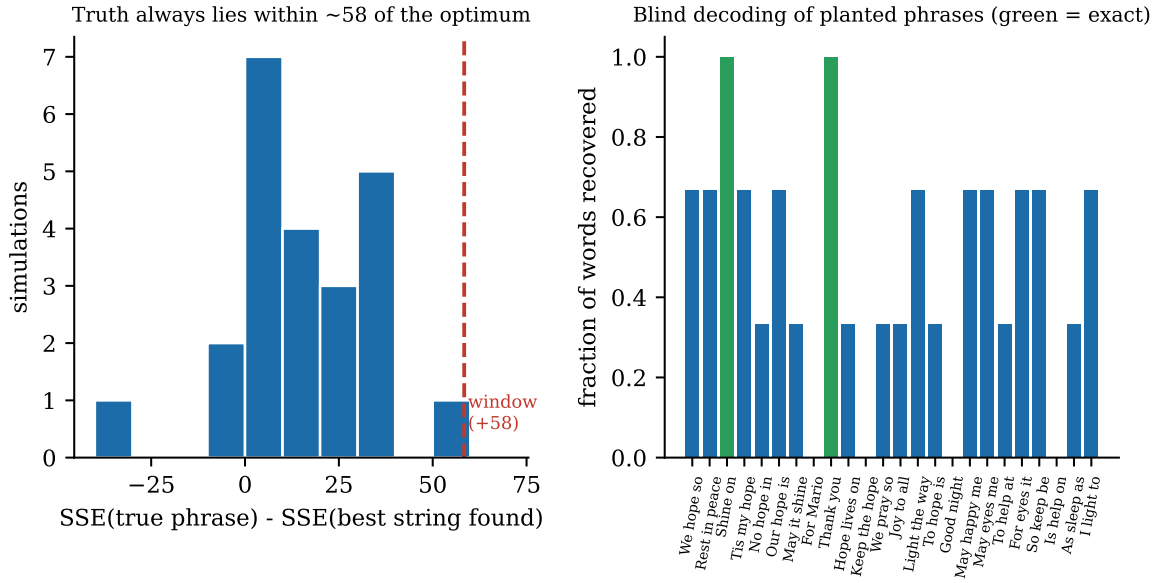


Figure 27: Calibration on synthetic plaques with known text: the gap between the true phrase and the best string found (left), and the words recovered per simulation (right).

10 The artist's font file: Adobe's 1990 PostScript Palatino

The family is settled. We then asked which *file* the artist used. Every digitisation of Palatino we could obtain went through the identical pipeline: line-1 recipe search, then line 2 calibrated on "hope" (Table 5 and Fig. 28).

- **URW Palladio**, the 1990s PostScript clone, in three generations: Ghostscript 5.10 (1998) and 6.0 (1999) from the GNU archive, and TeX Live's Palladio L.
- **TeX Gyre Pagella** and **URW P052**.
- **Apple's System 7.5.3 Palatino** (1996).
- **Palatino Linotype** (1998).
- **Adobe's own Type 1 Palatino** (`Palatino-BoldItalic 001.005`, dated 2 July 1990), from an *Adobe Type Basics* disc of 1994. This copy was supplied by the reader and is not redistributed.

Adobe fits best. Adobe's original fits best on both lines: line 1 731.3, line 2 311.2, total 1042.5 against 1085.1 for the best clone, an improvement of 43.

- **Recipe.** 192×160 canvas, 26 px, tracking +2, blur 0.30, gamma 1.4, at 4.41 times the noise floor, the closest any rigid model came.
- **Only Palatino benefits.** The other genuine Adobe fonts on the same disc (Times, New Century Schoolbook, Garamond, Zapf Chancery, Bookman) do *not* beat their clones. The Adobe advantage is specific to Palatino.
- **Unhinted rendering is consistently worse.** The original was rasterised *with* hinting, as Adobe Type Manager did in 1996.

The blind search now finds the sentence by itself. With Adobe's Palatino, the unconstrained dictionary search over 17.7k English words returns exactly **"We hope so"** as its single best string, in both settings (Fig. 29):

- line 1's typesetting: 362.1, ahead of "We hope he";
- hope-calibrated line 2: 311.2, ahead of "We hope as".

With every other font the top result had been a non-word such as "Wt hope he". On its own pixels, "so" is now the best of *all* endings, non-words included (91.4 vs. "nr" 92.7). We conclude that the artist most likely typeset the plaque with Adobe's PostScript Palatino Bold Italic, as found on Macintoshes with Adobe Type Manager in the mid-1990s.

One pixel the fonts cannot explain. At the bottom of the "W" the real plaque shows bare stone (row 19, column 8; value 8) between two bright points (Fig. 30). Every font predicts ink there, Adobe included (14.8). We tested and rejected global explanations:

- full 1990s TrueType hinting;
- sharpening after the resize;
- a drop shadow or emboss;
- 1996 Mac per-size advance widths;
- unhinted rendering.

Moving only the W by one artwork pixel improves line 2 markedly (311 → 281 with Adobe) but does not empty the hole. A second, similar hole sits between "Ete" and "rnal" on line 1. The most plausible explanation is the artist's hand: manual kerning or a one-pixel touch-up of the finished 32×32 texture, both common practice for tiny text.

digitisation	line 1	line 2	total	hole (19,8)
Palatino Bold Italic (Adobe Type 1, 1990)	731.3	311.2	1042.5	14.8
URW Palladio BI (Ghostscript 6.0, 1999)	754.9	330.2	1085.1	14.1
URW Palladio L BI (TeX Live)	754.9	330.2	1085.1	14.1
Palatino Bold Italic (URW P052)	756.9	329.0	1085.9	14.2
URW Palladio BI (Ghostscript 5.10, 1998)	755.1	332.2	1087.4	14.1
TeX Gyre Pagella BI	756.6	331.5	1088.1	14.0
Palatino Bold Italic (Apple 1996)	759.2	350.4	1109.5	13.8
Palatino Italic (Apple 1996)	809.1	329.3	1138.4	14.3
Palatino Linotype BI (1998)	793.8	367.3	1161.1	15.2
URW Palladio Italic (gs 5.10)	804.3	391.6	1195.9	14.1
Palatino Italic (Adobe Type 1)	802.9	395.7	1198.5	14.1

Table 5: Palatino digitisations under the identical pipeline (hinted rendering). Lower is better. The last column is the predicted value at the W hole; the real plaque shows 8.

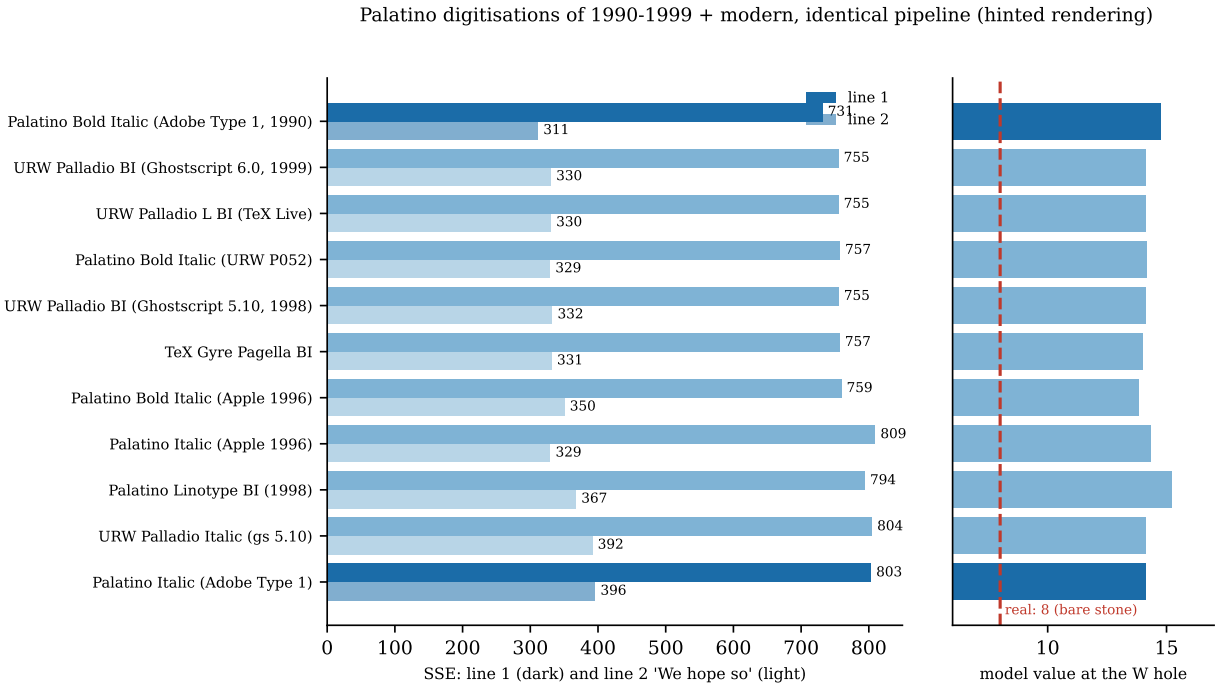


Figure 28: Line-1 and line-2 errors for each Palatino digitisation (left) and the predicted value at the W hole (right).

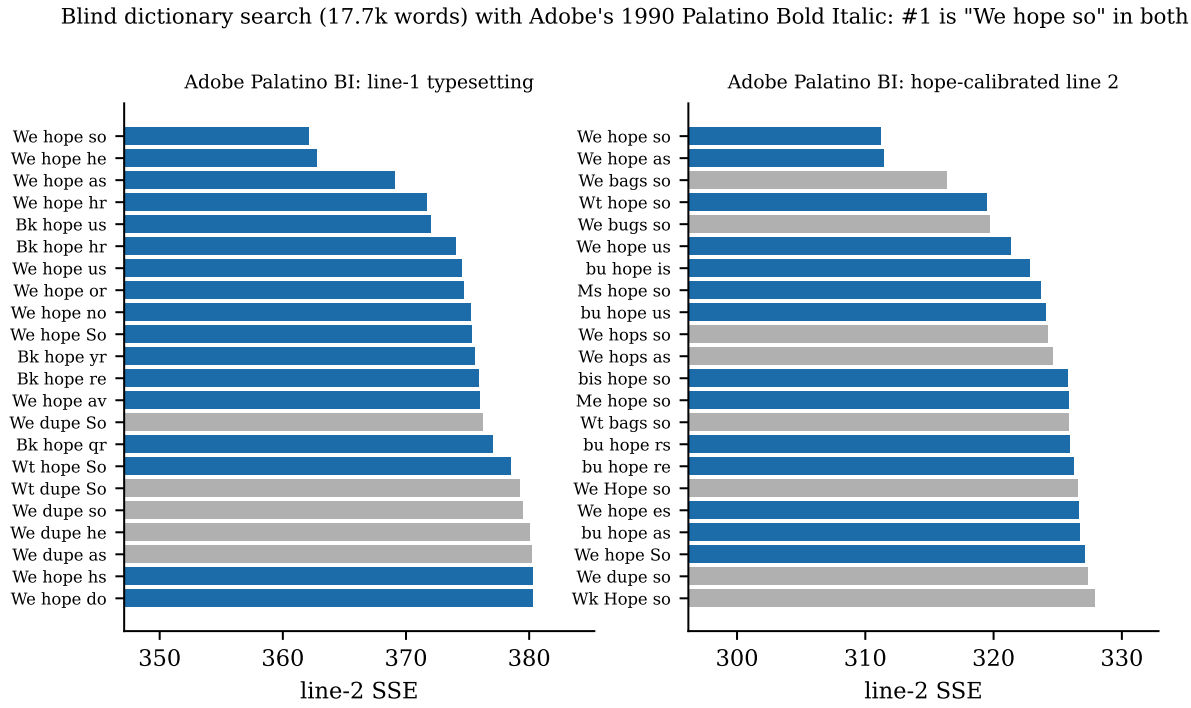


Figure 29: Blind dictionary search with Adobe's 1990 Palatino Bold Italic. "We hope so" is the single best string of the whole search, with line 1's typesetting (left) and with the hope-calibrated line 2 (right).

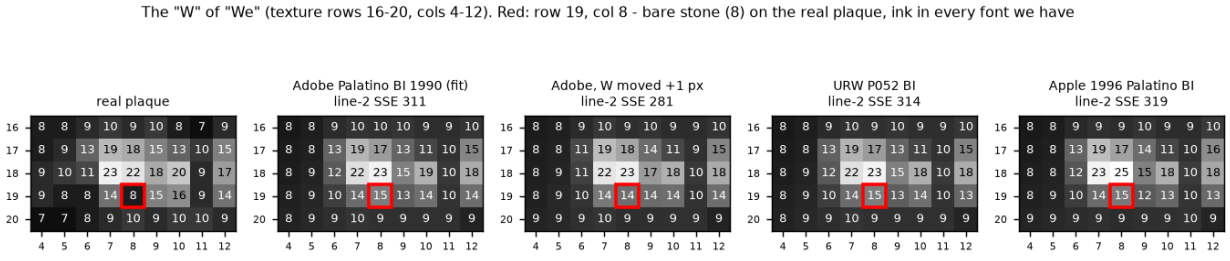


Figure 30: The W of "We", texture rows 16–20, columns 4–12, with grey levels printed. The red box marks the pixel that is bare stone on the real plaque but inked in every font.

11 Inverting the process: can the text be made visible?

The forward model maps a high-resolution artwork to the 32×32 texture, so it can also be run backwards. We solve for the 160×128 artwork whose rendering, after the fitted resize, blur, gamma, ink level and stone removal, reproduces the plaque.

Why priors are needed.

- **Without a prior the problem is wildly underdetermined.** The inverse reproduces the texture *exactly* (residual 0), including its stone noise, and looks like noise (Fig. 31, column 2).
- **Prior used.** We therefore add a sparsity and smoothness prior: most of the artwork is empty, and ink comes in strokes. Optionally the ink is also pushed to be black or white.
- **Setting its strength.** By the *discrepancy principle*: the leftover error must equal the measured noise floor of the two bands (331). The solver may then explain the ink, but not the noise.

Control plaques. A prior can draw letter-like strokes on its own, so we apply the *identical* inversion to synthetic plaques with known text.

- **Construction.** Every hypothesis gets the same smooth stone and the same real stone noise, taken from the clean rows 23–31. For "L is real 2401", row 2 keeps the real pixels because that claim says nothing about it.
- **Text is recoverable in principle.** On these plaques the inversion returns legible text ("Eternal Star", "L is real 2401", "A Secret star"). A correctly modelled plaque at this noise level could therefore be partly "read" after inversion.
- **The real plaque is less legible.** It inverts to letter-like blobs with the word structure "E ternal star / We hope so". The cause is not noise but the remaining model mismatch of Section 5: the real 1996 letter spacing and rasterising differ slightly from our fonts.

Similarity to the hypotheses. The correlation between the real inversion and each synthetic inversion, in the artwork space, is:

- row 1: **0.74** for "Eternal Star" vs. 0.29 for "L is real 2401" and 0.30 for "A Secret star";
- row 2: **0.76** for "We hope so" vs. 0.68 for "My hope is" and 0.44 for "lie here ~".

The inverted plaque looks like the inversion of "Eternal Star / We hope so" and clearly unlike the published alternatives. This is consistent with, and independent of, the forward scoring in the previous sections.

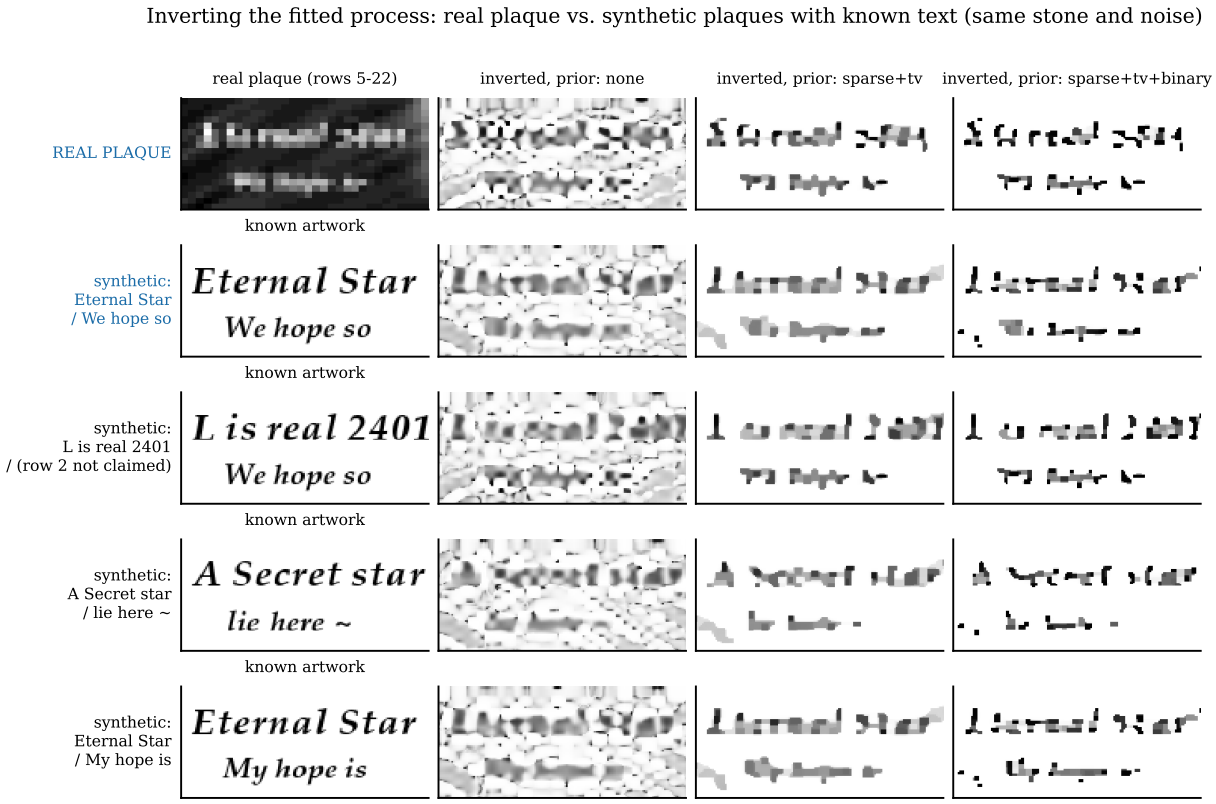


Figure 31: Inverting the fitted process. Top row: the real plaque and its inversion with three priors. Other rows: synthetic plaques with known artwork (left) on the same stone and noise, inverted identically. Column 2 (no prior) reproduces the pixels exactly but is meaningless. Columns 3–4 use the prior strength that leaves exactly the noise floor.

12 Second pass: the plain stone, every string and every era font

Readers asked three things: can the stone be taken out of the fit, does a non-word or a word outside our lists fit better, and was Palatino really the font available to the artist? We answered each directly.

The plain stone. Every pixel outside two boxes drawn around the visible text is stone and is kept unchanged (line 1: rows 8–12, columns 2–29; line 2: rows 16–20, columns 6–24; 235 of 1,024 pixels). Inside the boxes the stone is filled by Gaussian-process inpainting from all other pixels: an anisotropic kernel along the visible streaks (bottom left to top right; 10 px along, 4.5 px across) plus a quadratic trend (Fig. 32; `python -m plaquedecode stone`). The model is then scored as $\sum_i w_i\,(t_i - s_i - L\,c_i)^2$ with the stone s fixed and $w_i = 1/(0.15 + \mathrm{var}(s_i))$, so the stone can no longer absorb ink.

Same recipe, same words.

- The recipe search against the fixed stone returns exactly the Adobe recipe of Section 10 (26 px, 192×160, tracking +2, bilinear, blur 0.3, gamma 1.4; SSE 788.7). Apple's and URW's best recipes are unchanged as well (Table 2).
- Word 2: "Star" is the best of 12,547 words (Star 789, Stat 895, Slot 899, Sine 922). Among 149,699 entries of a large multilingual list it ranks 3 (Stai 760, Stor 766, Star 789, Slur 796): only the Italian "stai" and the Scandinavian "stor" come before it.
- Rival texts in the same font and process: "Eternal star" 881.7, "L is real 2401" 2031.3, "A Secret star" 2087.8.

Every string, not just words. We scored "Eternal " followed by *every* string of three or four letters (first letter in either case; 949,104 strings, four word spaces each) exactly, with linear glyph tiles under the frozen recipe. "Star" ranks 14; all strings ahead of it share its shape, "St/Sl" + a round letter + a short letter (Stoi 735, Stni 751, Stai 760, Slor 764, Stui 765, Stor 766). The third and fourth letters (a/o, r/i) are not separable

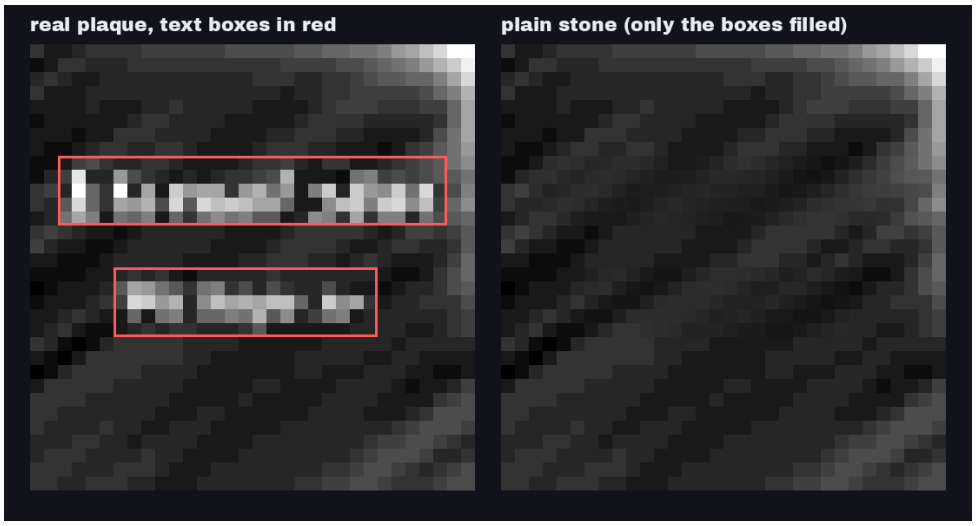


Figure 32: The text boxes (left) and the plain stone (right): only the 235 pixels inside the boxes are filled; every other pixel is the plaque itself.

at this resolution; the pixels force "St" + round + short letter, and "Star" is the only common word of that shape. On line 2 every slot was opened the same way:

- first word, all 36,558 strings of 1–3 letters: "We" ranks 8 behind W-shaped non-words ("Wi hope so" 220, "Wj hope so" 231, "Wf hope so" 240, "Wl hope so" 243);
- ending, all 36,559 strings: "so" ranks 4 ("We hope nr" 251, "We hope ur" 253, "We hope ar" 255, "We hope so" 262);
- middle word, all 456,976 four-letter strings: "hope" ranks 40 ("We bape so" 211, "We bope so" 218, "We bupe so" 222, "We baps so" 226).

Each slot lands in the top 0.01 % of all strings, and in each case the strings ahead are the same shape class (b and h are not separable at 3.5 px). The pixels fix the shapes; the language picks the words.

The first word, too. Because the recipe was fitted with "Eternal Star", we also opened the first word. Under the same frozen recipe and fixed stone, "X Star" was scored for about 245,000 dictionary entries (as written and capitalised) and for 2.1 million letter strings of 4–10 letters, built letter by letter by a blind beam search without any dictionary (each partial string is scored on the texture columns its letters already determine; 60,000 kept per length and line shift). The 1,237,405 candidates whose width lets "Star" land on its pixels were then scored exactly, with the whole line free to shift by up to two texture pixels (`python -m plaquedecode.firstword`). "Eternal" ranks **first**. The blind search finds it on its own; the closest rivals are one-letter non-word lookalikes ("Lternal" +6, "Eterual" +16, "Eternol" +26); the best other real word, "Sternal", fits 570 worse (rank 13,217).

Was Palatino available to the artist? We read every package of the IRIX 5.3 installation CD for the Indy (August 1995; SGI workstations were the N64 development platform). IRIX ships Adobe Times, Helvetica, Courier, Symbol and Utopia, Bitstream Charter and 14 SGI display faces, but no Palatino. Two details link Palatino to it anyway:

- SGI's online-documentation viewer expects `/usr/lib/DPS/outline/base/Palatino-BoldItalic` (symbolic links in `insight.sw`), together with New Century Schoolbook and Zapf Chancery.
- IRIX's printing package installs `/usr/sbin/typeoncall`, "TypeOnCall for IBM PC CD-ROM": it unlocks fonts from Adobe's PC *Type On Call* disc and converts their `.PFB` files into that directory.

That disc's `POBI____` metrics file is byte-identical to the one that accompanies the Adobe 1990 file used here. The Palatino Bold Italic an SGI artist could install in 1995 was therefore exactly this Adobe 001.005 face.

470 era faces under one process. We then collected every outline face an artist could plausibly have had in 1993–1996 from operating systems and bundled applications: IRIX 5.3 (including Showcase), Adobe Illustrator 5.5 for IRIX (221 Adobe/ITC faces), the 1990 Adobe Type 1 disc, CorelDRAW 6/7 (the Bitstream library), Microsoft Office 4.3, Font Pack 2 and Bitstream TrueType packs, Apple KanjiTalk 7.5, System 7.5.3 and Mac OS 9, Photoshop 3 (ATM), FreeHand 5, and Windows 3.1/95 (Japanese). All 470 italic faces ran through the same recipe search in three tiers (a quick grid for all, the font-comparison grid for the top 60,

#	face	source	SSE	recipe
1	Palatino Bold Italic 001.005	Adobe Type 1, 1990	788.7	26 px, 192×160, γ 1.4
2	Palatino Bold Italic 003.001	Adobe Type 1, 1999	808.3	32 px, 192×192, γ 1.4
3	Palatino Bold Italic	Apple System 7.5.3 / KanjiTalk 7.5	811.4	23 px, 160×128, γ 1.8
4	Palatino Bold Italic 3.5a3	Apple Mac OS 9	811.4	23 px, 160×128, γ 1.8
5	P052 Bold Italic	URW clone	814.0	32 px, 192×192, γ 1.4
6	Palladio L Bold Italic	URW clone (TeX Live)	821.4	23 px, 160×128, γ 1.8
7	URW Palladio Bold Italic	URW clone (Ghostscript 6.0)	821.4	23 px, 160×128, γ 1.8
8	URW Palladio Bold Italic	URW clone (Ghostscript 5.10)	822.2	23 px, 160×128, γ 1.8
9	Zapf Calligraphic 801 Bold Italic	Bitstream	847.1	26 px, 192×160, γ 1.4
10	Palatino Linotype Bold Italic	Linotype, 1998	859.3	32 px, 192×192, γ 1.4
11	Book Antiqua Bold Italic	MS TrueType Font Pack, 1992	884.5	27 px, 192×160, γ 1.4
12	Zapf Calligr BT Italic	CorelDRAW 6 (Bitstream)	896.4	31 px, 192×192, γ 1.8
13	Schneidler Blk BT Black Italic	CorelDRAW 6 (Bitstream)	926.4	36 px, 256×192, γ 1.8
14	Dutch 811 BT Bold Italic	CorelDRAW 6 (Bitstream)	946.1	25 px, 192×160, γ 1.4
15	Lapidary 333 BT Bold Italic	CorelDRAW 6 (Bitstream)	950.6	32 px, 192×160, γ 1.4
16	Book Antiqua Italic	MS Office 4.3	953.6	27 px, 192×160, γ 1.4
17	Century Expd BT Bold Italic	CorelDRAW 6 (Bitstream)	954.3	31 px, 256×192, γ 1.4
18	Bodoni Bold BT Bold Italic	CorelDRAW 6 (Bitstream)	961.2	35 px, 256×192, γ 1.0
19	Palatino Italic 001.005	Adobe Type 1, 1990	973.2	27 px, 192×160, γ 1.4
20	C059 Bold Italic (New Century Schoolbook)	URW clone	980.7	21 px, 160×128, γ 1.8

Table 6: Line 1 against the plain stone, best recipe per face (470 faces; identical files merged).

the full grid for the top 8). The best is Adobe's 1990 Palatino Bold Italic (788.7); the twelve best distinct faces are all Palatino digitisations or close Palatino derivatives (Table 6 and Fig. 33). The best face of any other design is Schneidler Blk BT Black Italic at 926; IRIX's own Times Bold Italic is at 1,116.

What is left. With the stone fixed, the remaining large errors sit on the *top* strokes of line 1 (E top arm, S top, the r arm) and in the W of line 2, where the model inks pixels the plaque shows as bare stone. To explain them by the stone it would have to be darker than any stone pixel on the plaque; per-letter vertical shifts do not beat a noise-only control. Horizontal per-letter shifts do; Section 13 follows this up like a designer would.



Figure 33: Line 1: the real plaque, each font's best fit and its error map (brighter = worse).

13 Third pass: thinking like the designer

So far every letter sat exactly where the font put it. A real artist in 1996 typed the text, looked at it, nudged letters, maybe added an effect, and only then shrank the artwork to 32×32. We now let the model do the same things and ask, for every step, one question: *does it help more on the real plaque than on pure noise?*

How we decide that a step is real. Any extra freedom lowers the error a little, even on random data. So for every step we also build *fake plaques*: our best model plus the real plaque's leftover error, shuffled to random positions. A fake looks like the plaque but contains no hidden structure. We run exactly the same search on several fakes. A step counts as real only if it helps the real plaque clearly more than it helps every fake.

Step 1: hand kerning. Each letter may move left or right by whole pixels of the artwork, the way an artist nudges letters with the arrow keys (one artwork pixel is a sixth of a texture pixel). Line 1 improves from 788.7 to 694.9 (fakes: 0–66), line 2 from 261.5 to 201.0 (fakes: 0–11). Both gains are far above the fakes: the letters really were placed slightly differently than the font places them. In line 2 the nudges pull "so"

designer step	gain (real)	gain (fakes)	verdict
text layer blended with the stone as "Screen" (or "Normal")	14	11–18	not real
text layer blended as "Lighten"	much worse (1366)		ruled out
engraved / drop shadow	24	9–36	not real
emboss / bevel	34	7–32	borderline
outline around the letters	1	0–11	no
sharpen after shrinking	−24	−39 to 3	no (worse)
move a whole line up or down	−165	−178 to −156	no (worse)
artwork 190×160 instead of 192×160, re-kerned	41	0	**real**

Table 7: Designer steps on top of the hand-kerned two-line canvas. "Gain" is how much the error drops.

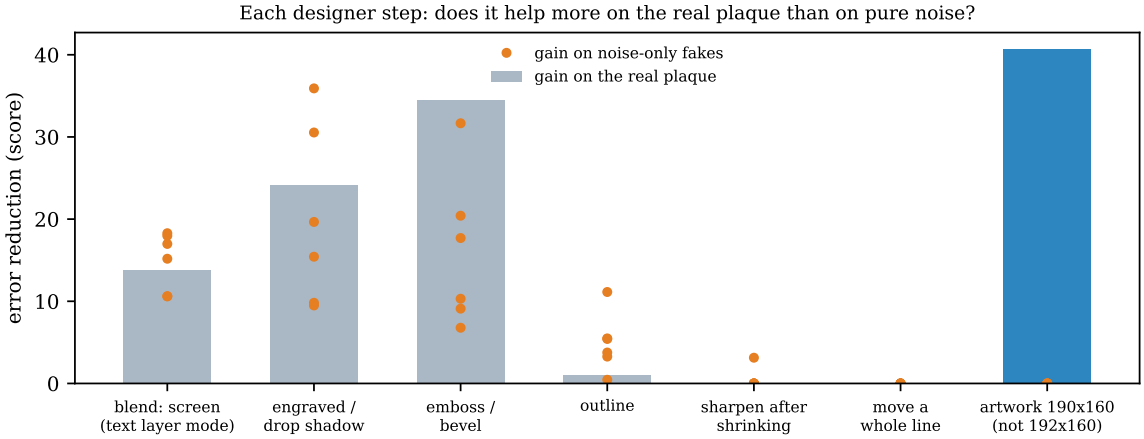


Figure 34: Blue bars: gain on the real plaque; orange dots: gain on noise-only fakes. Only the slightly narrower artwork (and the hand kerning before it) clearly beats noise.

slightly to the left and move the W slightly right. We also searched every one of the 2,000 best recipes with hand kerning allowed: the winner is the same recipe (26 px, 192×160, tracking +2, bilinear), so kerning does not change the font, the size or the process, only the letter positions.

Step 2: one canvas for both lines. Both lines are now typed on the same artwork, kerned together and shrunk together, with one ink level per line: 1045.5 before kerning, 894.1 after.

Step 3: everything else a designer could have done. Table 7 and Fig. 34 list each step with its gain on the real plaque and on the fakes. The one step that clearly survives is the artwork size. Our recipes assumed artwork exactly 6×5 times the texture (192×160). Allowing any artwork size between 176 and 208 pixels wide and 136 and 184 high, the best is **190×160**: about 1 % narrower, error 894.6 → 853.9; on the fakes this freedom bought nothing. The text was set a little tighter than the font sets it, which the hand kerning had already hinted at.

The best version. Our best reconstruction is therefore: Adobe Palatino Bold Italic (1990), line 1 at 26 px and line 2 at 19 px, typed on one 190×160 artwork, letters nudged by at most a third of a texture pixel, shrunk to 32×32 with a bilinear filter, light blur and gamma 1.4, on the extracted stone (Fig. 35). The fourth pass (Section 14) re-checks line 2's font and refines this version slightly (Fig. 1). Inside the two text boxes the model is on average 1.1 grey levels away from the plaque (out of 31); 83 % of the text pixels are within 2 levels.

What no designer step explains. The same seven pixels stay wrong in every version (yellow in Fig. 35): the top arm of the E, two pixels on top of the S, two pixels around the first "t", the arm of the r, and the gap at the bottom of the W. At each of them the plaque shows bare stone while every typeset version predicts faint grey ink. No font, size, filter, blend mode or effect removes exactly these pixels without breaking others.

The simplest explanation is that the artist touched up the finished 32×32 texture by hand and cleaned away a few faint grey edge pixels. This cannot be modelled from the font, and it changes nothing about the reading.

What the clean pixels have in common. To test the touch-up idea we looked at *every* pixel inside the text boxes where the best model expects only faint ink (2–8 grey levels): 71 pixels. On the plaque, 15 of them are exactly bare stone, including the seven above. Sorting all 71 by where they sit in the letter shapes (Table 8 and Fig. 36):

where the faint pixel sits	pixels	bare stone on the plaque
in a gap between two letters or strokes	22	7 (32 %)
at the tip of a horizontal stroke	8	4 (50 %)
on the top edge of a stroke	8	1 (12 %)
on the bottom edge of a stroke	20	1 (5 %)
elsewhere	13	2 (15 %)

Table 8: Faint-ink pixels of the best model and how many of them are bare stone on the real plaque.

- The clean pixels are the ones that would make letters run together (around the first “t”, inside the W) or blur the tips of strokes (the arms of E and r). Faint pixels on the bottom edges of strokes almost always keep their grey.
- Whether faint ink shows does not depend on how bright the stone underneath is (correlation -0.08), so no blend mode explains it; and because most faint pixels keep their ink, no global threshold or filter does either.
- 12 of the 15 lie in the upper half of “Eternal Star”.

This is the signature of a legibility clean-up of the finished texture: gaps between letters opened and fuzzy stroke tips trimmed so that the tiny text reads more crisply. It is ordinary hand work on a small texture and the one part of the plaque that no font or setting can reproduce.

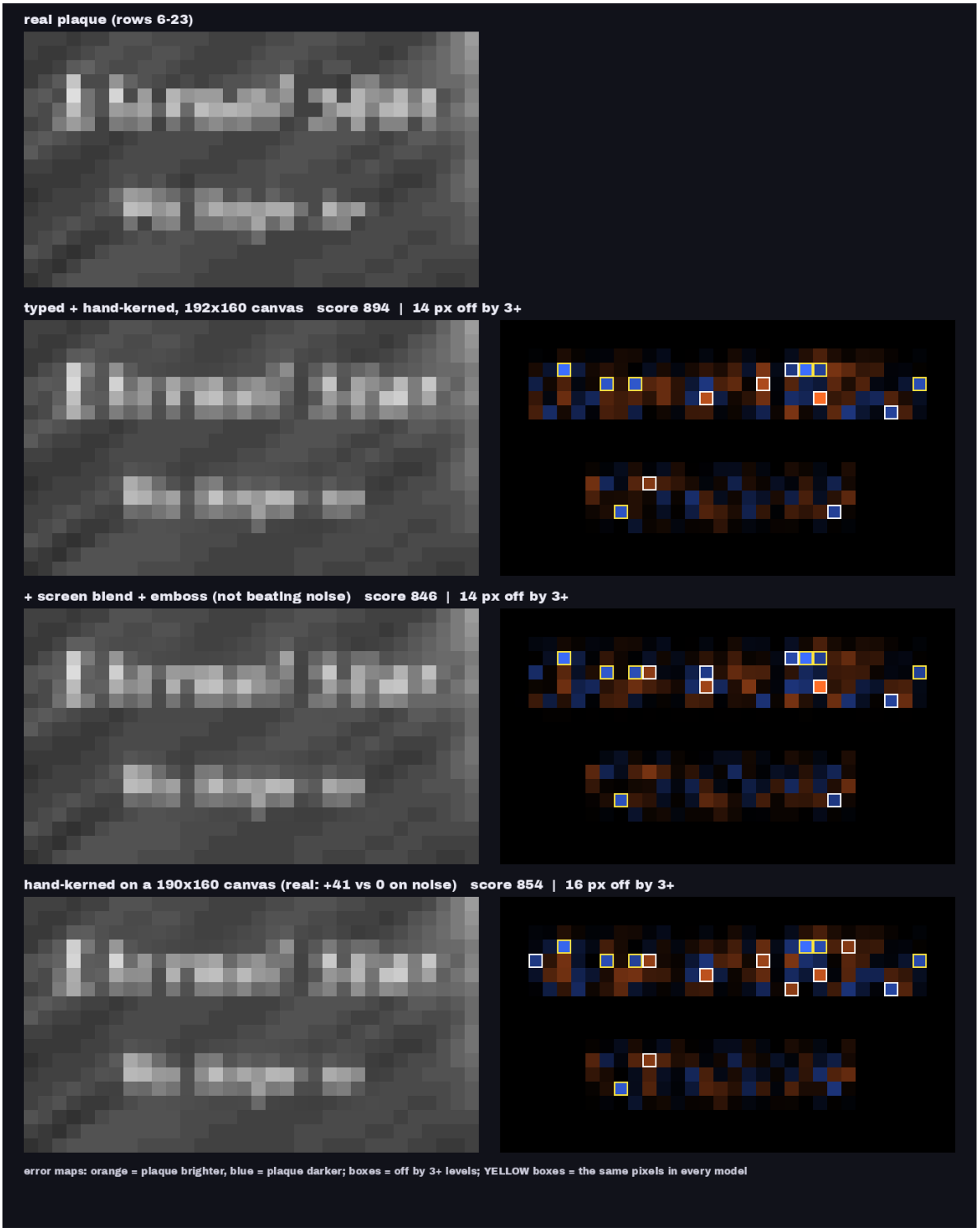


Figure 35: Both lines: the real plaque, the hand-kerned model, the model with the (not significant) blend and emboss steps, and the hand-kerned model on the 190×160 artwork. Right: error maps. Yellow boxes mark the pixels that stay wrong in every model.

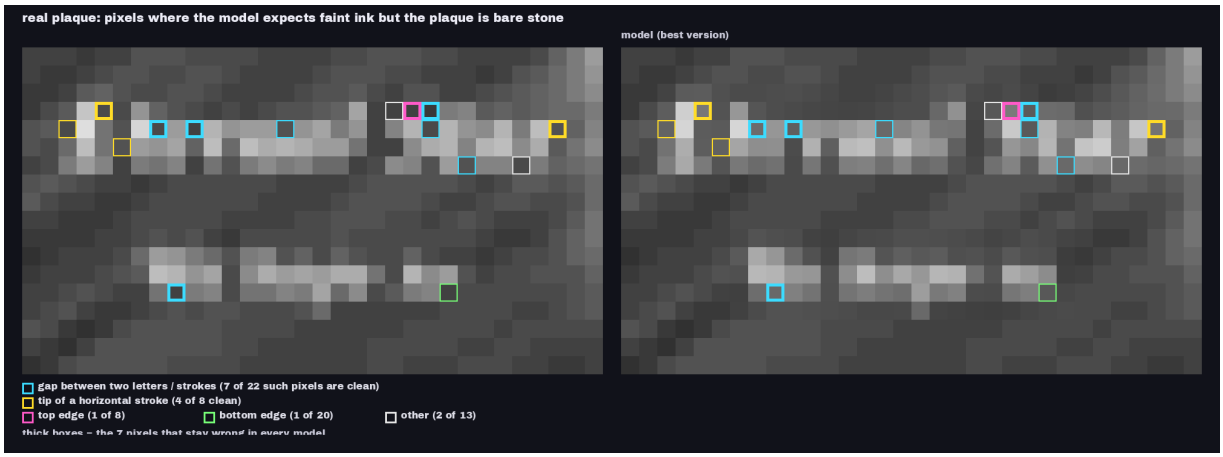


Figure 36: Real plaque (left) and best model (right). Boxes: pixels where the model expects faint ink but the plaque is bare stone, coloured by where they sit (cyan: gap between letters; yellow: tip of a stroke; magenta: top edge; green: bottom edge). Thick boxes: the seven pixels that stay wrong in every model.

14 Fourth pass: is line 2 in the same font? And the final fit

Every font tournament so far was decided on line 1. Line 2 was allowed its own size, spacing and kerning, but only among Palatino faces (Section 8). A designer could have set the small second line in a different typeface, so we tested that directly (`python -m plaquedecode.line2font`).

Step 1: every era font on line 2 alone. For this test we use a more flexible model than the rigid recipes: each font is drawn large and smooth, then placed, scaled, stretched, slanted, spaced, blurred and gamma-corrected with *continuous* settings that are fitted by gradient descent on the GPU. Gamma is limited to 0.5–2 and the ink level to what a 5-bit texture can hold: without these limits, picture fonts and blobby display faces "win" with absurd settings. We fitted 3,396 faces to line 2 alone: the era fonts of Section 12 plus our whole 1993–96 font library (CorelDRAW, Bitstream, Illustrator for IRIX, Office, KanjiTalk, Mac OS and more). The surprise: the six weights of **Raleigh** (an upright Bitstream serif from CorelDRAW) took six of the top seven places (error 150–158), ahead of Palatino Bold Italic (167–175). But every Raleigh fit slanted the upright font by about 14° to imitate an italic.

Step 2: the same model on both lines. If the flexible model simply liked Raleigh's shapes, Raleigh would also do well on line 1. It does not (Fig. 37, left): on "Eternal Star" the four Palatino Bold Italic files score 641–654 and beat the best wrong text ("Eternal Stat") by about 93, while every Raleigh scores 1,114–1,262 and often loses to wrong texts. So the model can tell fonts apart, and line 1 is unmistakably Palatino. Line 2, however, carries far less information (error without any text: 4,027 against 9,086 for line 1), and there the Raleigh lead stays (Fig. 37, middle).

Step 3: drawing the letters the way 1996 software did. The flexible model draws every glyph large and smooth. A paint program in 1996 drew them at the real pixel size with *hinting*: the font rasteriser snaps stems and letter heights to whole pixels, and the plaque was made with hinting (Section 10). At 19 px, the size of line 2, hinting changes letter shapes far more than at 26 px. So we repeated the test with the release pipeline's own rendering: hinted, true pixel size, line 1's canvas, filter, blur, gamma and ink level locked, and line 2 free in size, spacing, kerning, word space, position and even a paint-program skew of up to 0.3 (Fig. 37, right):

- **Palatino Bold Italic wins** (Mac OS 9 file 229.6, KanjiTalk 236.4, Adobe 263.1), with no skew. With Adobe's file the best setting is again 19 px, no kerning and whole-pixel advances, exactly the "hope" calibration.
- Raleigh Demibold comes close (232.6) only when skewed by 0.25, and even then it explains "We hops so" *better* than "We hope so": it is not reading the text, it is filling blobs.
- Every other face is far behind (Officina Serif 271.8, Cooper 273.1, the other Raleigh weights 295–558).

The Raleigh lead of step 1 was a side effect of drawing the letters smooth instead of hinted. **Line 2 is set in Palatino Bold Italic, like line 1.**

Step 4: the final fit. Before the hand kerning, Apple's TrueType Palatino looked slightly better than Adobe's on line 2. To check whether line 2 was typed with a different *copy* of Palatino, we took the designer's best version (Section 13) and replaced line 2 with each Palatino Bold Italic file, re-searched its size, spacing and position on the shared artwork, re-nudged every letter of both lines and re-searched the artwork size. Once the letters are hand-kerned, Adobe's file wins on line 2 as well: **850.95** against 856.68 for the Mac OS 9 and Apple files. The best fit is

- one artwork of 190×160 pixels for both lines, shrunk to 32×32 with a bilinear filter, blur 0.3 and gamma 1.4, on the extracted stone; ink level 43 (line 1) and 48 (line 2);
- line 1: Adobe Palatino Bold Italic (1990), 26 px, tracking +2, pair kerning, whole-pixel advances;
- line 2: the same file at 19 px, no extra tracking, no kerning, whole-pixel advances, slightly wider word spaces;
- every letter nudged by −2 to +1 artwork pixels by hand (at most a third of a texture pixel).

It improves the version of Section 13 from 853.89 to 850.95, with the same quality inside the text (on average 1.1 grey levels from the plaque, 83 % of the text pixels within 2 levels). Blending the text as "Screen" would lower the error to 841.3, but that gain (9.6) is smaller than what blend modes gain on noise-only fakes (10.6–18.3), so we do not claim it. This final fit and the artwork it implies are shown in Fig. 1 at the start of the report; it is also the title image of the repository.

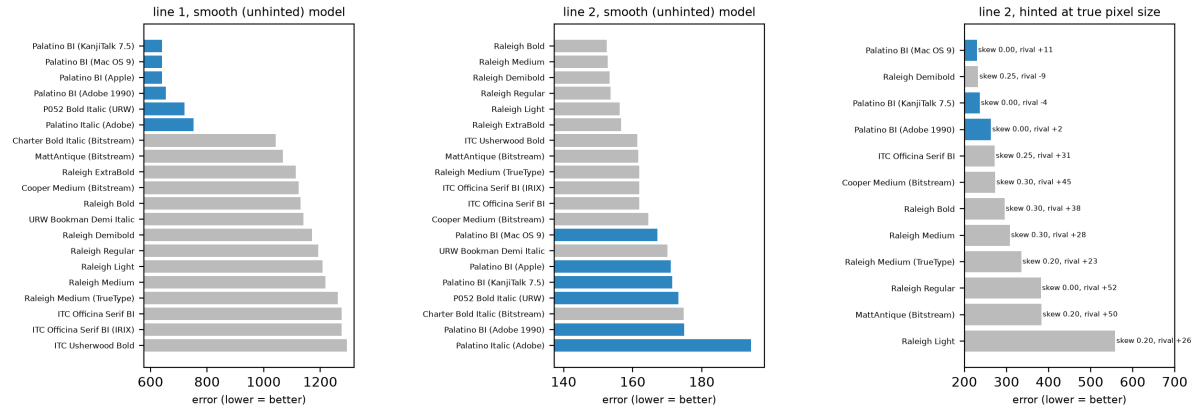


Figure 37: Is line 2 in another font? Blue: Palatino files. Left and middle: the flexible (smooth, unhinted) model on line 1 and on line 2. Right: line 2 drawn hinted at its true pixel size, with the best skew and the margin over the best wrong text (negative: a wrong text fits better).

15 Comparison with published readings

Every claim was scored with the same model. Each claimed line 1 received its *own* exhaustive recipe search. Line 2 was scored jointly under that recipe, and also under the "Eternal Star" recipe. The results are in Table 9, Fig. 38 and Fig. 39.

"L is real 2401". The legend puts the whole phrase on row 1 and makes no claim about row 2, so we scored it that way. Row 1 received its own exhaustive recipe search, and row 2 was not scored.

- **Row 1 fit.** Even so, "L is real 2401" fits row 1 about twice as badly as "Eternal Star": 1492 (Apple) against 759. The "2041" spelling does the same (1561).
- **Why.** The texture shows a long, evenly inked first word with a tall capital at the start and no gap after it. "L is" predicts a short capital L and a two-letter word separated by spaces, and the digits predict wide, cap-height shapes where the texture has italic lower-case letters (Fig. 39).
- **Row 2.** Its text is left unexplained by this reading.

Our results agree with the 1998 Nintendo letter that the legend has no basis in the texture [4].

"A Secret star / lie here ~" (Nodd). Nodd correctly identified "star" and a three-part line 2. However, "A Secret" fits line 1 far worse than "Eternal", and "lie here ~" has no descender where the texture has one.

"Eternal Star" (Super Mario Wiki). Confirmed. It is the best line 1 of all, and its "heavily serifed font" [1] is Palatino Italic.

"My hope is". This reading gets "hope" right, but it predicts a second descender, the "y", which is absent (Fig. 14). It is also clearly worse than "We hope __" in both fonts (Table 9).

line 1	line 2	proposed by	line-1 SSE		line-2 SSE	
			Apple	URW	Apple	URW
“L is real 2401”	(not claimed)	1990s rumour (‘Luigi is real’), whole phrase on row 1	1492	1414	–	–
“L is real 2041”	(not claimed)	variant spelling, row 1 (e.g. MobyGames trivia)	1561	1437	–	–
“A Secret star”	“lie here ~”	Daniel Nodd (No64DD), Nintendo Life	1604	1460	1126	1320
“L is real star”	(not claimed)	reader comment, Nintendo Life	1421	1378	–	–
“Eternal Star”	(not claimed)	Super Mario Wiki (‘more likely’)	759	757	–	–
“Eternal Star”	“My hope is”	proposed reading (two descenders + ascender)	759	757	511	459
“Eternal Star”	“my hope is”	same, lower case	759	757	489	475
“Eternal Star”	“lie here ~”	Nodd’s line 2 under Eternal Star	759	757	695	689
“Eternal Star”	“We hope so”	this work	759	757	373	396
“Eternal star”	“We hope so”	this work (lower-case s)	765	755	554	565

Table 9: Published and proposed readings. Lower is better. Line-1 noise floor 165.7. The line-2 pixel-compatible window is roughly 344–402.

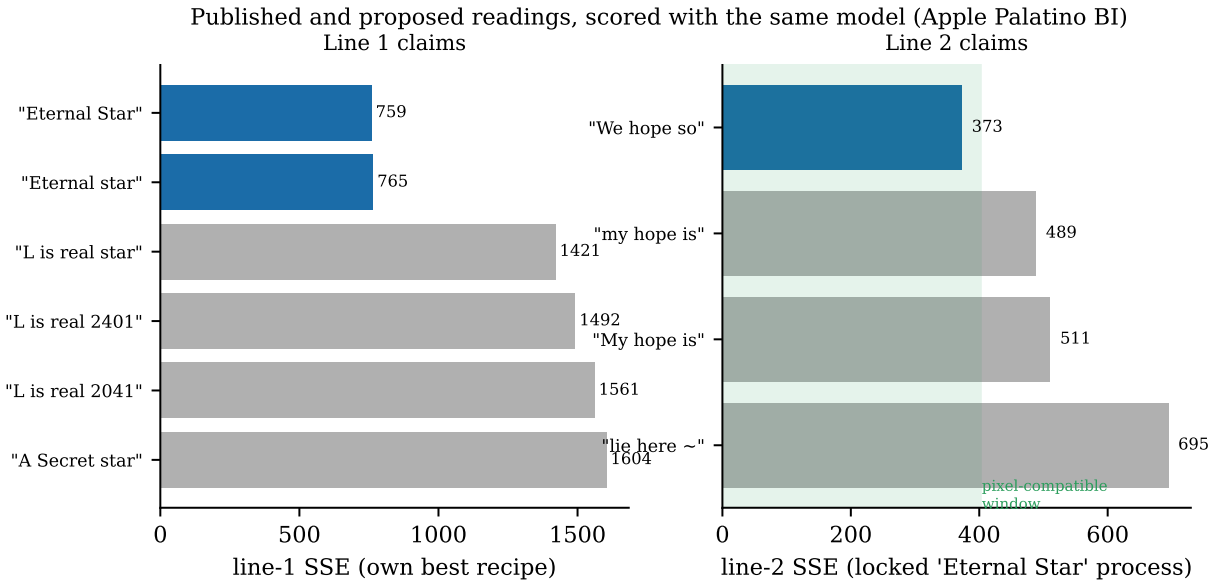


Figure 38: Line-1 and line-2 errors of the claims. The green band is the line-2 pixel-compatible window.

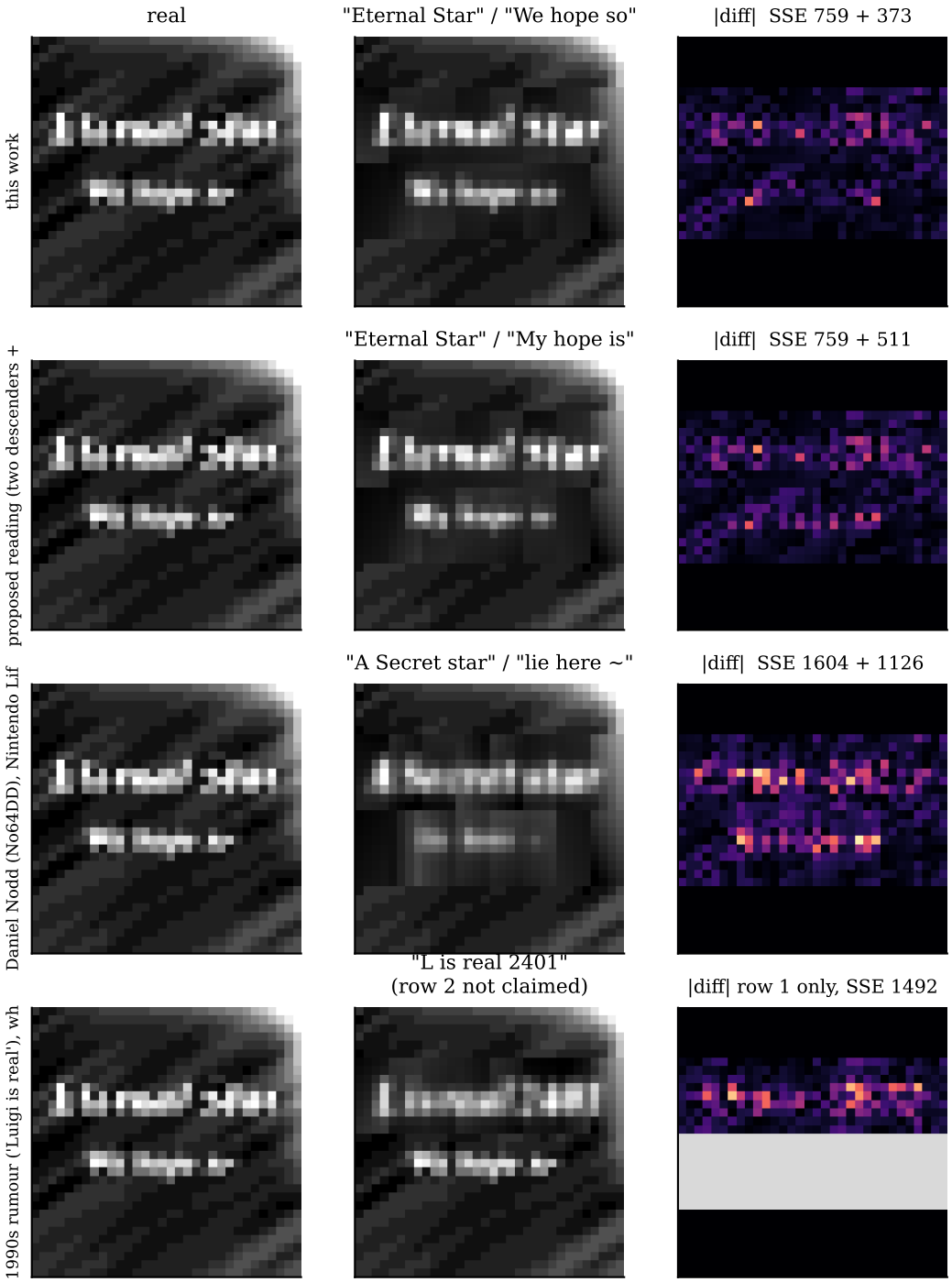


Figure 39: Whole-plaque reconstructions of four readings: real plaque, model rendering and absolute difference.

16 Does the reading make sense in Japanese?

This section is interpretation, not evidence. The pixels decide the reading; here we only ask whether the decoded text is something a Japanese artist in 1995–96 would plausibly have written. The plaque texture is identical in the Japanese and all later versions, so it was written in English from the start, in Japan.

A memorial-plaque pattern. Japanese monuments and commemorative plaques (記念碑) very often consist of a name or theme followed by a wish, for example 「永遠の平和」(“eternal peace”) followed by を願って (“wishing for it”). Our reading has exactly this two-part shape: **Eternal Star** names the statue (永遠の星 or 永遠のスター), and **We hope so** is the wish (そうであってほしい, politely そう願っています). An elegant bold-italic book face, like an engraving, suits such a plaque.

“We hope so” is a direct rendering of a Japanese thought.

- “I think so” and “I hope so” are among the first set phrases of Japanese school English; a Japanese writer composing a short English line would naturally reach for this ready-made block.
- English “so” works like Japanese そう: it points back to what was said before. On its own, “We hope so” is an odd sentence for a plaque; it only makes sense together with the title above it: *eternal star, and we hope it is so*. A native English writer would more likely have chosen something like “May it shine forever”; the slightly unusual phrasing fits an author translating his own Japanese thought quite literally.
- Japanese leaves out the subject, so the writer had to choose one. On a public plaque the collective voice (私たち, 我々), the castle’s residents or the development team, is the natural choice, which gives “We”.
- The tone is a modest wish rather than a claim, close to the literary blessing そうあれかし (“may it be so”).

“Eternal Star” as a title. 永遠 rendered as “Eternal” is a typical title word in Japanese-made English, and Japanese decorative English usually capitalises such titles, which slightly favours the capital “S” that the pixels alone could not decide. In the game the Power Stars (パワースター) protect the castle; a star statue wishing that its star lasts forever fits the story. Since スター in Japanese also means a star performer, the line may carry a second, playful meaning, the team’s wish that their new 3D Mario stays a star; we note this only as a possibility.

The pixel look-alikes do not make sense. The pixels also allow close shape rivals such as “We hope or”, “We hope us” or “He hope so” (Section 12). None of them is a plausible rendering of a Japanese wish, and none would make sense on a plaque. “We hope so” is the only candidate that is both compatible with the pixels and natural for the author.

Our best guess at the intended meaning.

永遠の星——そうであってほしい（そう願っています）
An eternal star, and we hope it truly is.

17 Limitations

- **Rasteriser.** The exact 1996 rasteriser (hinting, letter spacing) is unknown. This is the source of the remaining gap to flexible fits, about 4.6 times the noise floor.
- **Stone model.** The stone under the letters is modelled, not known.
- **Two-letter ending.** Under line 1’s typesetting the ending cannot be resolved by the pixels. After the hope calibration “so” is the best real ending, but its margin over “or”, “us” and “he” is small, and meaning still supports the choice.
- **Case.** Capital vs. lower-case “S” in “Star” is undecided: line 1 alone cannot separate them, and the joint two-line test points in opposite directions for the two fonts. Whether the final word is “so” or “So” cannot be decided either.
- **Vocabulary.** The dictionary searches are only as complete as their vocabulary. Very rare words, names and non-English text were not covered exhaustively.
- **Separate line fits.** The process is fitted on line 1 and then shared with line 2. In one joint test (URW), a slightly different process fitted line 2 much better (283) at a small cost on line 1. A fully joint two-line recipe search could tighten the model further. It would not change the reading: it was run with “We hope so”.

- **No source material.** We did not use original source art or leaked material. Such material, or a statement by the original artist, would be the only way to reach certainty.

18 Conclusion

A physically motivated forward model, searched exhaustively and calibrated on synthetic plaques, shows that the plaque was made by typing two centred lines in **Palatino Bold Italic**, most likely Adobe's 1990 Type 1 version: with that file the best artwork is 192×160 at 26 px (line 1), resized to 32×32 with a bilinear filter, lightly blurred and gamma-adjusted (1.4); the Apple and URW Palatinos independently converge on 160×128 at 23 px and gamma 1.8. Line 2 is set at 17–18 px in the same process. A second pass against the extracted plain stone, an exhaustive test of every 3–4-letter string and a tournament of 470 era faces (Section 12) leave all of this unchanged; the Palatino an SGI workstation could install in 1995 was exactly this Adobe file. Treated as a designer's workflow (Section 13), the letters were nudged by hand on a slightly narrower artwork (190×160); blend modes, shadows, emboss, outlines and sharpening do not beat noise, and the last handful of wrong pixels look like touch-ups on the finished texture. Line 2 is set in the same font as line 1: a flexible model briefly preferred an artificially slanted Raleigh, but drawn hinted at its true size Palatino Bold Italic wins again, and in the final fit (Section 14, error 850.95) Adobe's file is best on both lines. Read as English written by a Japanese artist (Section 16), the text is a natural memorial-plaque wish, 永遠の星——そうであってほしい: an eternal star, and we hope it truly is.

Our confidence in each word:

word	evidence	confidence
Eternal	every font, process and test	very high
Star / star	rank 1 of 27,122; rivals with their own recipes lose; case undecided (the two fonts disagree)	high (word), open (case)
hope	rank 1 of 12,556; appears unprompted in every global search; its letters calibrate line 2	high
We	best English word of the required shape; best real word on held-out pixels after the hope calibration; "By", "We'll" and "Me" fail on the whole line	high
so	two small letters are certain; best real ending after the hope calibration and the only one that completes the sentence; margin to "or", "us" and "he" is small	moderate

Running the fitted process backwards gives an independent visual check. The inverted plaque resembles the inversion of "Eternal Star / We hope so" far more than that of "L is real 2401" or "A Secret star / lie here ~" (Section 11).

The most likely text of the plaque is ***"Eternal Star — We hope so"***. That is a light-hearted line, in keeping with the 1998 description of the plaque as a developers' joke, and certainly not "L is real 2401".

A Reproducing the results

The code is released as the Python package `plaquedecode` (MIT licence). No ROM, texture or commercial font is distributed.

```
pip install -r requirements.txt
python -m plaquedecode extract --rom "Super Mario 64 (USA).z64"     # -> texture.npy (local)
# put P052-BoldItalic.otf (free, URW base35) into fonts/
R=results/recipes_P052-BoldItalic.json
python -m plaquedecode recipe   --font P052-BoldItalic.otf --stage full    # line-1 process
python -m plaquedecode word2    --recipes $R                               # word 2 of line 1
python -m plaquedecode line2    --recipes $R --mode dp                     # dictionary search
python -m plaquedecode line2    --recipes $R --mode combo --phrases curated
python -m plaquedecode candidates --recipes $R --line2 results/line2_dp_P052-BoldItalic.json
python -m plaquedecode hopefit  --recipes $R --line2 results/line2_dp_P052-BoldItalic.json \
                                --fonts2 P052-BoldItalic.otf
python -m plaquedecode simulate --recipes $R --line2 results/line2_dp_P052-BoldItalic.json
python -m plaquedecode claims   --font P052-BoldItalic.otf --recipes $R
python -m plaquedecode invert   --recipes $R --hopefit results/hopefit_PalatinoBoldItalic.json \
                                --line2 results/line2_dp_hopefit_P052line2.json
python -m plaquedecode figures  --out report/figures                       # + report/tables/
# fourth pass (fixed stone: PLAQUE_STONE=stone/stone_box.npy PLAQUE_STONE_VAR=stone/stone_box_var.npy)
```

```
python -m plaquedecode.firstword --recipes results/stone_recipes_Adobe.json \
        --out results/stone_firstword_Adobe.json                    # every first word
python -m plaquedecode.line2font screen --fonts-list FONTS.txt    # every face on line 2 alone
python -m plaquedecode.line2font both   --fonts-list SHORT.txt    # both lines + wrong-text controls
python -m plaquedecode.line2font hinted --fonts-list SHORT.txt    # hinted, true pixel size
python -m plaquedecode.line2font best   --fonts-list PALATINO.txt # final fit -> results/bestfit.json
python -m plaquedecode.line2font hero;  python -m plaquedecode.line2font figure
```

The searches take minutes to an hour on a consumer GPU. All result files used for this report are included in `results/`, so the figures and tables can be regenerated without a GPU.

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